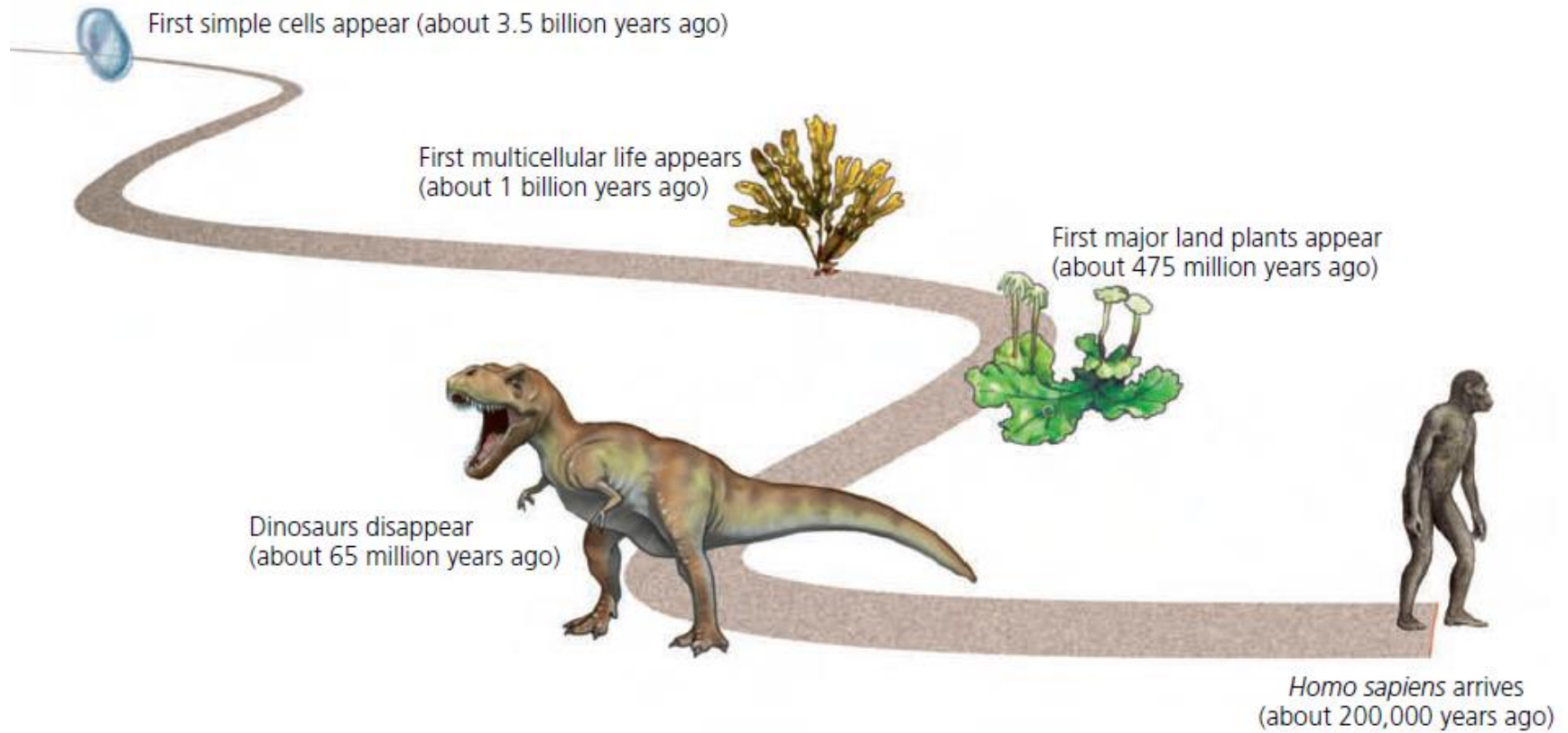


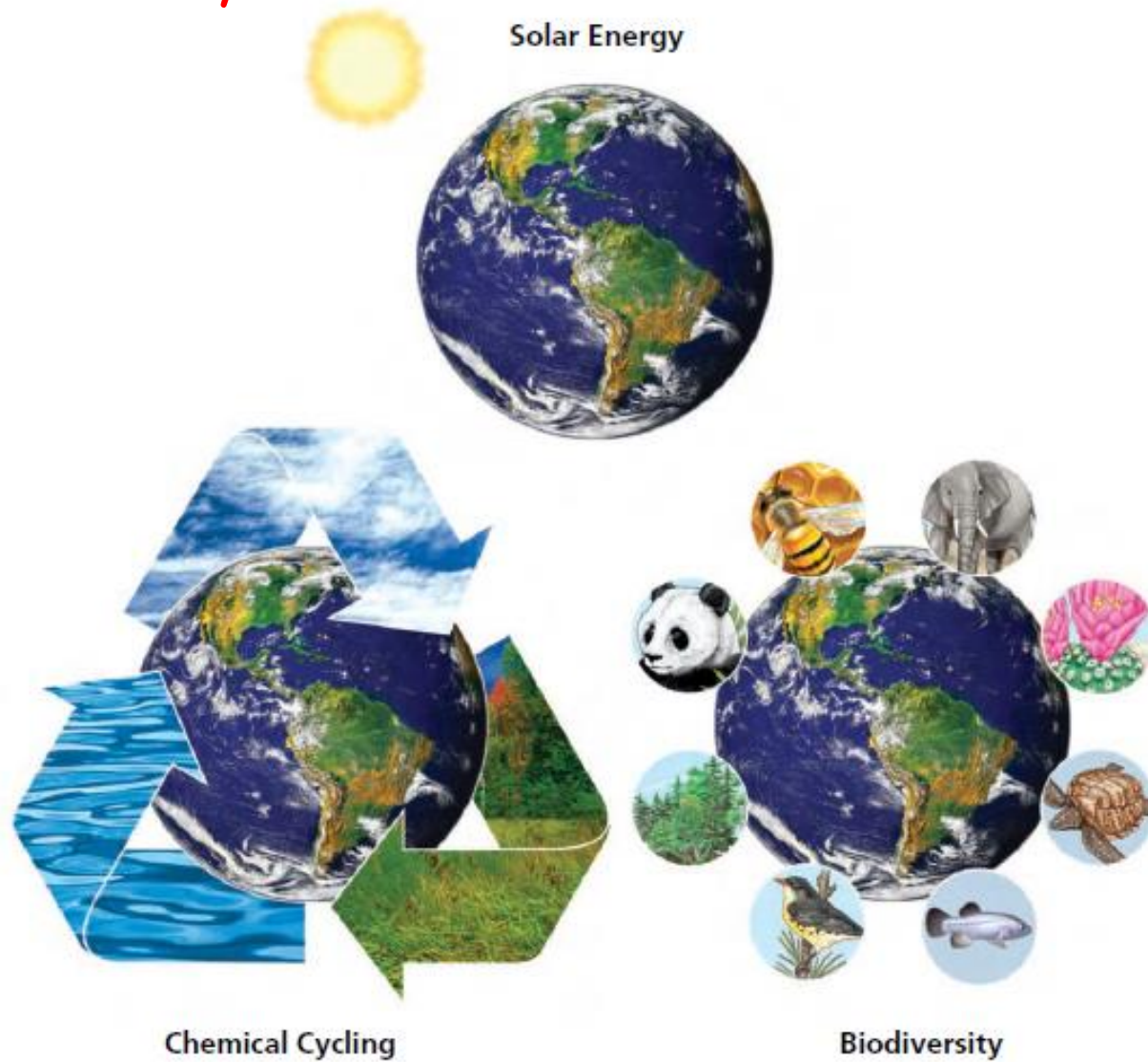
Biodiversity



Sustainability

- Considering the billions of years that life has existed on the earth, our species has been around for less than the blink of an eye
- Within only a few hundred years, we have taken over most of the earth to support our basic needs and rapidly growing wants. But in the process, we have degraded much of the earth.

Sustainability



Sustainability

There are three overarching themes relating to the long-term sustainability of life on this planet

- *solar energy- rely on the sun*
- *Biodiversity-promote multiple options for life*
- *chemical cycling-reduce wastage*

Biodiversity

- Importance
- types
- mega-biodiversity
- Species interaction
- Extinct, endemic, endangered and rare species;

Biodiversity

- **Biological diversity**, or **biodiversity**, is the variability among living organisms from all sources including *inter alia*, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part.

Types of biodiversity

Types of Biodiversity

- *Species diversity*
- *Genetic diversity*
- *Ecosystem diversity*
- *Functional diversity*

Species diversity

- **Species diversity** is the variety of species per unit area and this include both the **species richness** (number of species present) and their **relative abundance** (how many of each species per unit area)

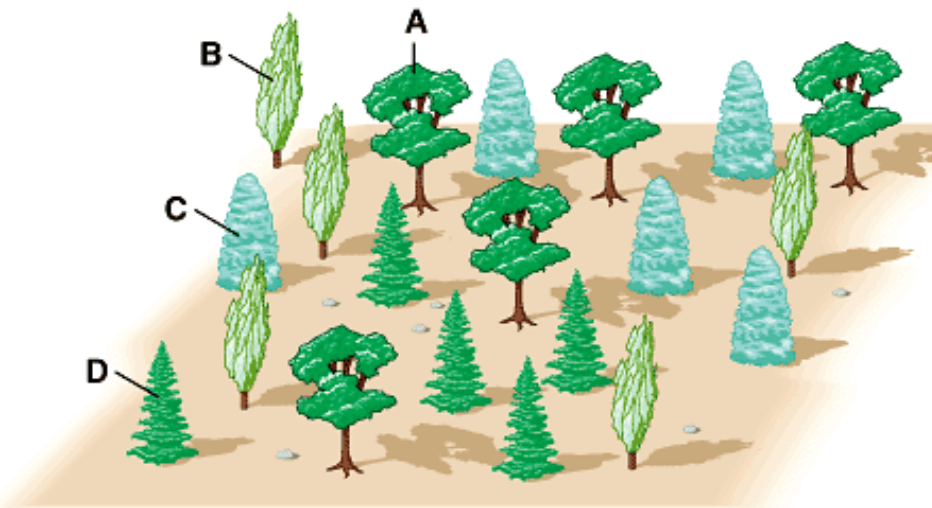
Species diversity

- We do not know how many species there are on the earth.
- Estimates range from 8 million to 100 million.
- So far, biologists have identified about 1.9 million

Species Diversity



Species Diversity



Community 1

A: 25% B: 25% C: 25% D: 25%



Community 2

A: 80% B: 5% C: 5% D: 10%

In these images we can see that the two communities are composed of the *same* species richness, but they *differ* in regards to species abundance:

Genetic diversity

Genes are the basic units of hereditary information transmitted from one generation to other

The genes found in **organisms** (e.g. Plants, animals, humans) can form enormous number of combinations each of which gives rise to some variability.

When the genes within the same **species** (e.g. cats, lions) show different versions due to new combinations, it is called genetic diversity.

Genetic diversity



Genetic Diversity in Human



Ecosystem diversity

- ***Variation in the habitat*** in which species live and interact
- ***Different types*** of forest, grass lands, wet lands, water bodies like streams, lakes and oceans, mountains...



Functional diversity

the variety of processes such as energy flow and matter cycling that occur within ecosystems as species interact with one another in food chains and food webs.

Importance of biodiversity

Importance of biodiversity

- The earth's biodiversity is a vital part of the natural capital that helps keep us alive
- Biodiversity also plays critical roles in preserving
 - a) quality of the air and water
 - b) maintaining the fertility of topsoil
 - c) Decomposing and recycling waste
 - d) controlling populations of species that humans consider to be pests.

Biodiversity helps to sustain life on the earth.

Importance of Biodiversity

Direct Values

- Direct values are concerned with the enjoyment or satisfaction received *directly* from biological resources.
- Relatively *easily observed and measured* by assigning prices to them.
- There are two types including *Consumptive* and *Productive* use.

Importance of Biodiversity – Direct values

Consumptive use

- Direct values where the biodiversity products can be harvested and consumed directly, like *food, fuel, milk, fiber etc.*
- Humans are mostly dependent on plants and animal resources for *dietary requirements*. A major share of the food comes from domesticated crops and animals.
- A significant percentage of world population depends upon plants or plant extracts for *medicines*.
- Forests have been used since ages for *fuel wood*. Firewood collected by individual is not normally marketed, but is directly consumed by *tribes and local villagers*.

Direct Values – Consumptive Use



Importance of Biodiversity – Directive values

- **Productive values** are commercially usable values where the product is marketed and sold (Productive value is assigned to products that are ***commercially harvested and marketed***).
- **Animal products** like tusks of elephants, musk from musk deer, silk from silkworm, wool from sheep... all of which are traded in the market.
- Several industries dependent upon the productive use values of biodiversity, like ***paper industry, plywood industry, textile industry, leather industry, pearl industry etc.***



Direct Values – Productive Use



Importance of Biodiversity-indirect values

- These are **indirect values** of ecosystem functions.
- **Carbon fixing** through photosynthesis, which provides the **support system** for species.
- **Pollination**
- Maintaining **nutrient cycles**
- Soil **production and protection** from erosion
- **Absorbing and decomposing** pollutants, organic wastes, pesticides, air and water pollutants
- **Regulating climate**
- Preserving recreational, aesthetic, cultural and historic values of **natural environment**

Importance of Biodiversity

- **Non-consumptive values** deal primarily with the functions of ecosystems.
- *Do not normally appear* in national accounting systems, but they may **outweigh** direct values when they are computed.
- Non-consumptive values include
 - Social values**
 - Aesthetic values**
 - Ethical values**

Importance of Biodiversity

- **Social values** are associated with social life, customs, religious aspects of the people. Many of the plants are considered holy and sacred, like tulsi, peepal, lotus etc. Leaves, flowers and fruits are used in worship.
- Many animals like cow, snake, bull, peacock etc. also have significant place in the religious arena.



Importance of Biodiversity

- **Aesthetic value**: Biodiversity is responsible for the *beauty* of a landscape.
- People from far and wide spend a lot of time and money to visit *wilderness* area where they can enjoy the aesthetic value of biodiversity.
- This type of tourism is known as *eco-tourism*. Eco-tourism is estimated to generate a huge revenue annually that gives the aesthetic value of biodiversity.



Importance of Biodiversity

- **Ethical value**: Ethical or moral value is based on the principle of “*Live and let others live*”.
- All forms of life have *equal right* to exist on earth and human being is *only one small part* of the earth’s great family of species.
- *People take pleasure in hunting on animals (?!)*
- In addition, *degrading and polluting* the environment also classified as *unethical actions*, which should be avoided through proper *education and awareness*.

Importance of Biodiversity

- **Option value:** These values include the potentials of biodiversity that are presently unknown and need to be explored.
- Thus option value is the value of knowing that there are biological resources existing on this biosphere that may one day prove to be an effective option for something important in the future.

Endemic Species

Endemic Species

- Species that are found in only one area are called **endemic species** and are especially vulnerable to extinction.



Bicolored frog
Western Ghat, India



Lemur
Madagascar



Giant Panda
China

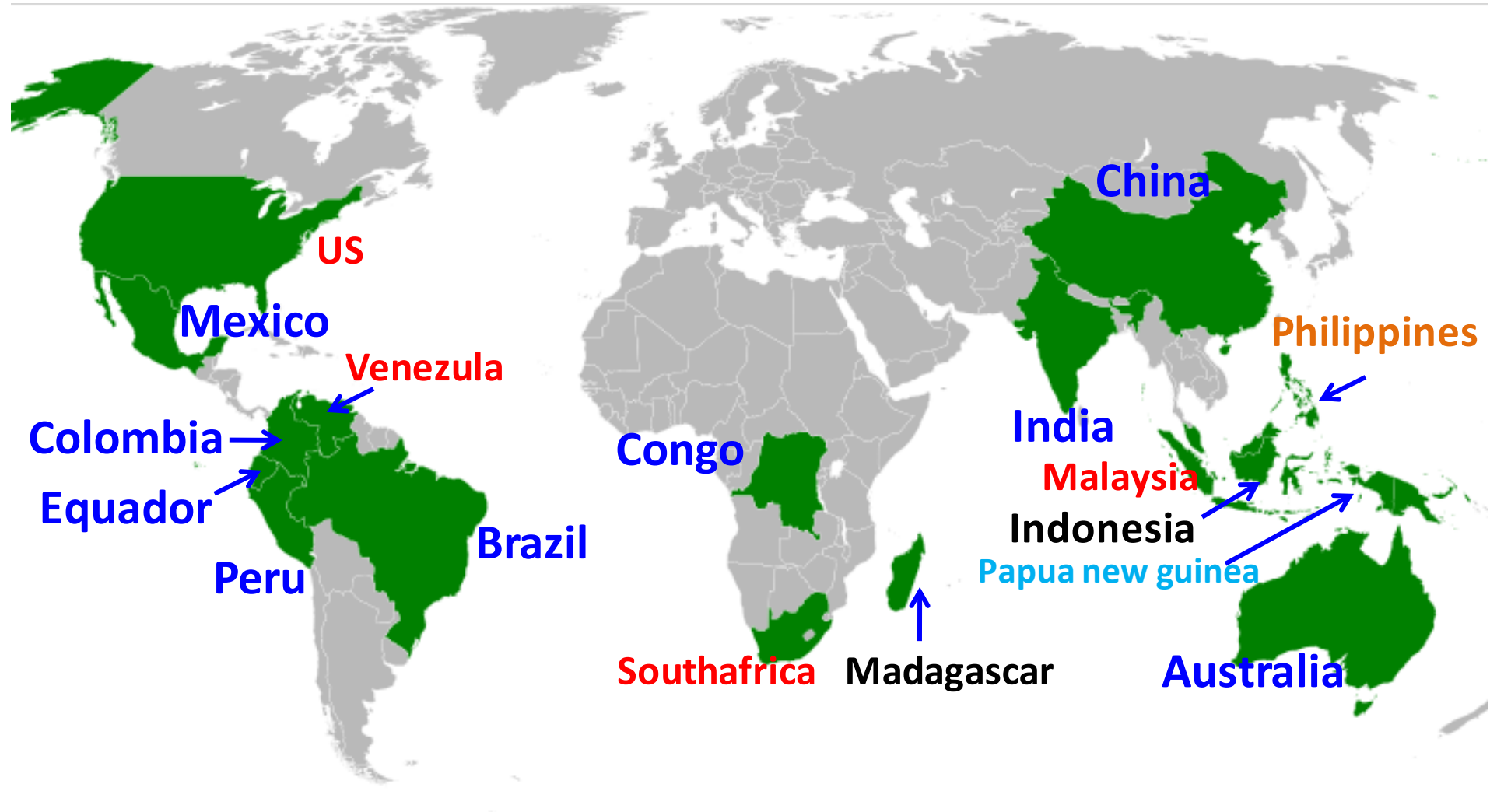
Megadiversity

Megadiversity

- ***Megadiverse country*** refers to any one of a group of nations that harbour the *majority of earth's species and high number of endemic species*.
- Megadiversity means exhibiting ***great diversity***.
- Main criteria for megadiverse countries is ***endemism*** at the level of species, genera and families.
- (***Endemic species*** are plants or animals which are native to and restricted to a particular geographical region)
- A megadiverse country must have at least ***5,000 species*** of endemic plants and must border marine ecosystems.

Megadiversity

There are **17** Megadiverse countries.



Species Interaction

Species Interaction

Ecologists identify five basic types of interactions between species as they share limited resources such as food, shelter, and space:

- **Interspecific competition**
- **Predation**
- **Parasitism**
- **Mutualism**
- **Commensalism**

Species Interaction

- **Interspecific competition**

occurs when members of two or more species interact to gain access to the same limited resources such as food, water, light, and space.



Species Interaction

- Interspecific competition

Each species plays a role in its ecosystem called its *ecological niche*

What is ecological niche

It is a species' way of life in a community and includes everything that affects its survival and reproduction, such as how much water and sunlight it needs, how much space it requires, what it feeds on, what feeds on it, and the temperatures it can tolerate.

Species Interaction

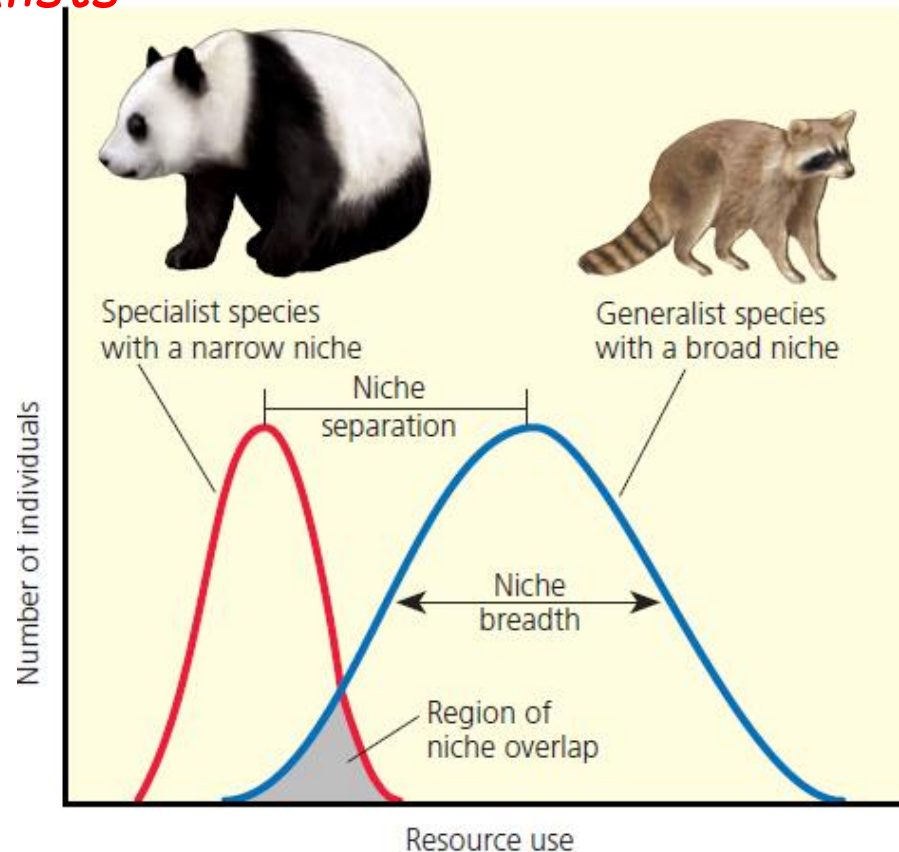
- Interspecific competition

Each species plays a role in its ecosystem called its *ecological niche*

Scientists use the *niches* of species to classify them broadly as *generalists or specialists*

Generalist species have broad niches

specialist species occupy narrow niches



Species Interaction

- Interspecific competition

Each species plays a role in its ecosystem called its *ecological niche*

Some species are specialists with a narrow niche and some are generalists with a broad niche.

When two species compete with one another for the same resources such as food, light, or space, their *niches overlap*.

The greater this overlap, the more intense their competition for key resources.

Species Interaction

Predation: Occurs when a member of one species (the predator) feeds directly on all or part of a living organism (the prey).



Species Interaction

- **Parasitism**

Parasitism occurs when one species (the *parasite*) feeds on another organism (the *host*), usually by living on or inside the host. In this relationship, the parasite benefits and the host is often harmed but not immediately killed.

Examples: Bedbugs, tapeworms, mosquito



Species Interaction

- **Parasitism**

From the host's point of view, parasites are harmful.

But from the population perspective, parasites can promote biodiversity by contributing to species richness,

in some cases, they help to keep the populations of their hosts in check.

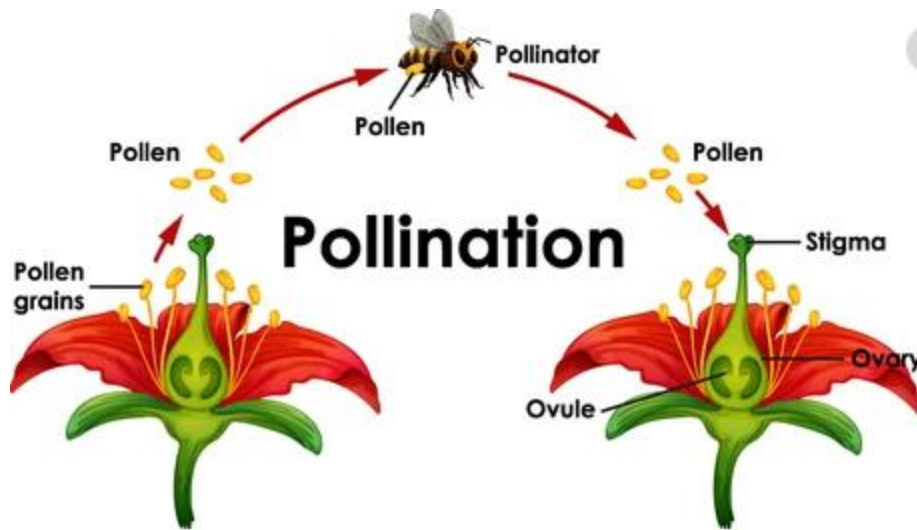
Species Interaction

- **Mutualism**

In **mutualism**, two species behave in ways that benefit both by providing each with food, shelter, or some other resource.

pollination of a flower by species such as honeybees

butterflies that feed on the flower's nectar



Species Interaction

- **Mutualism**



(a) Oxpeckers and black rhinoceros

mutualistic relationships that combine *nutrition* and *protection*.

The birds remove and eat parasites and pests (such as ticks and flies) from the animal's body and often make noises warning the larger animals when predators are approaching.

Species Interaction

- Commensalism

Commensalism is an interaction that benefits one species but has little, if any, beneficial or harmful effect on the other.



birds benefit by nesting in trees,
generally without harming them.

Species Interaction

- Commensalism



plants called *epiphytes* (such as certain types of orchids and bromeliads), which attach themselves to the trunks or branches of large trees in tropical and subtropical forests

Species Extinction

Species Extinction

The balance between formation of new species and extinction of existing species determines the earth's **biodiversity**.

A species is said to be **extinct** when it is not seen in the wild for 50 years at a stretch e.g. passenger pigeon.

Affect on ecosystem

Biological extinction of certain important species can **weaken or break** some of the connections in the **ecosystem**

Biological extinction can lead to **secondary extinctions (co-extinction)** of species with strong connections to species that have already gone extinct.

Species Extinction

Species extinction can happen naturally

During most of the 3.5 billion years that life has existed on the earth, there has been a natural, low rate of species extinction known as the **background extinction rate**.

The extinction of many species in a relatively short period of geologic time is called a **mass extinction**.

earth has experienced mass extinctions. After each mass extinction, biodiversity eventually returned to equal or higher levels, but each recovery required millions of years.

Species Extinction

Species extinction can happen naturally

Reasons for species extinction

Major climate change

Volcanic eruption

a large asteroid hit the planet

Species Extinction

Some Human Activities are Causing Extinctions

Extinction is a natural process but it has accelerated as human populations have spread over most of the globe,

We consume huge quantities of resources and creating large and growing ecological footprints

filling in wetlands or converting grasslands and forests to crop fields and urban areas.

Human activities have also polluted and disturbed almost half of the surface waters

Species Extinction

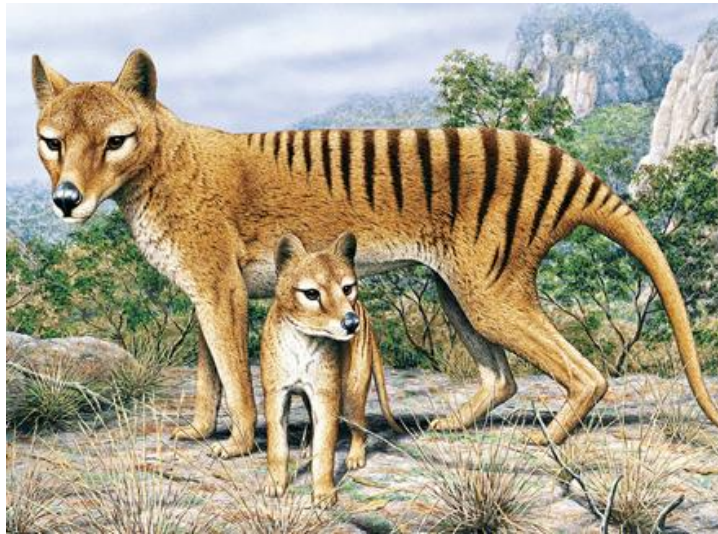
Some Human Activities are Causing Extinctions

- Current annual rate of species extinction is up to 1,000 times the historical background rate of about 0.0001%.
- biodiversity losses are likely to increase sharply during the next 50-100 years because of the projected growth of the human population and its growing **use of resources per person**,
- we may be **limiting the long-term recovery** of biodiversity by reducing the rate of speciation for some species. In other words, we are also creating a speciation crisis.

Some of the extinct species



Passenger Pigeon
Extinction in 1914



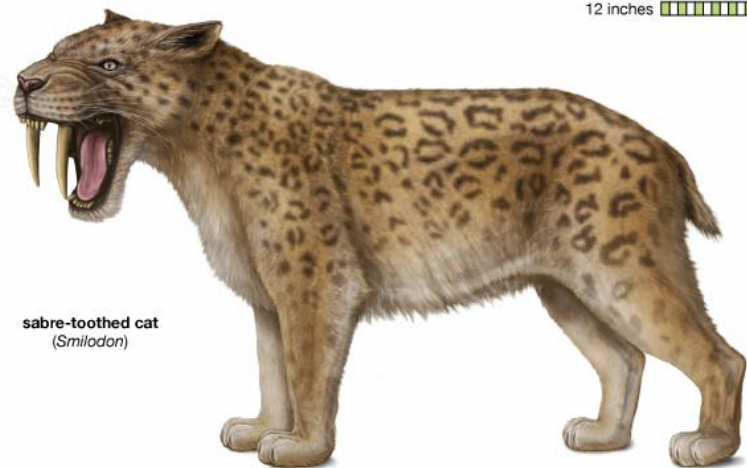
Tasmanian Tiger
Extinction in 1936



Dodo
Extinction in 1662



Mammoth
Extinction 10,000 – 4000 BC



Sabre-toothed cat
Extinction about 10,000 BC

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Passenger Pigeon Extinction



- These birds were good to eat,
- Their feathers made good pillows,
- Their bones were widely used for fertilizer.

North America's passenger pigeon extincted because

- habitat loss
- uncontrolled commercial hunting
- They are easy to kill

Endangered Species

Endangered Species

Biologists classify species that are heading toward biological extinction as

Endangered

Threatened

rare

An **endangered species** has so few individual survivors that the species could soon become extinct.

A **threatened species** (also known as a *vulnerable species*) still has enough remaining individuals to survive in the short term, but because of declining numbers, it is likely to become endangered in the near future.

Endangered Species

Biologists classify species that are heading toward biological extinction as

Endangered

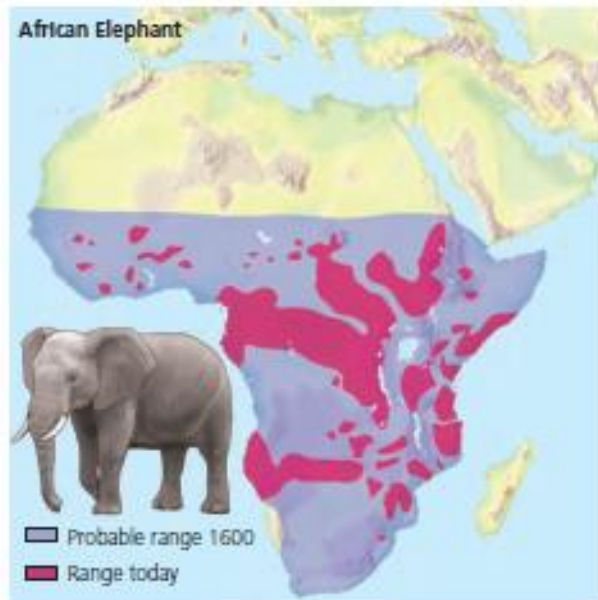
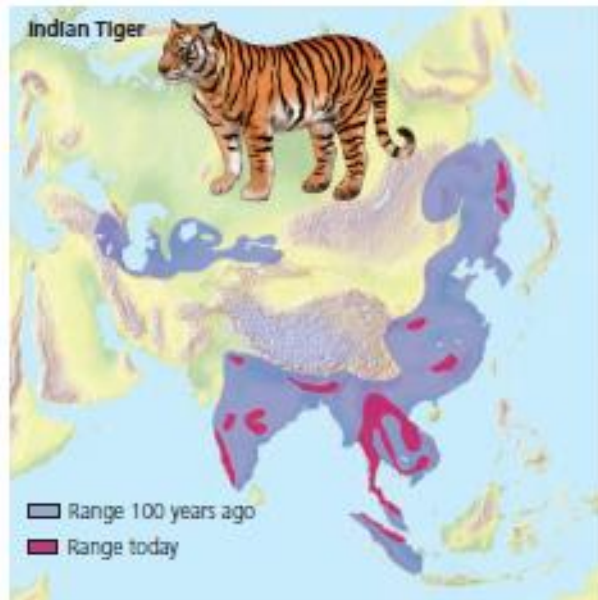
Threatened

rare

Species which are not endangered or vulnerable at present, but are at a risk are categorized as **rare species**.

they are usually endemic. Sometimes they are thinly scattered over a more extensive area.

Endangered Species



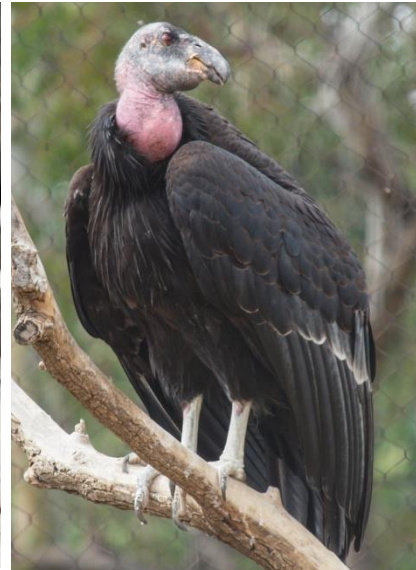
Some of the endangered species



Sumatran Tiger
<60 in Indonesia



Mexican Grey Wolf
< 60 in New Mexico



California condor
172 in US



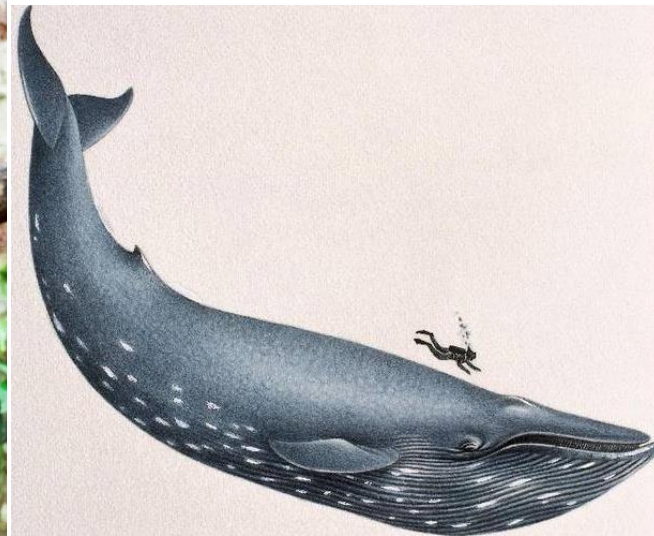
Whooping Crane
210 in US



Indian Bustard



Orangutan



Blue Whale

Endemic Species

- They exist on islands and in other unique areas, especially in tropical rain forests where most species have highly specialized roles.

Extinction of Endemic Species is because

- Habitat loss
- changing environmental conditions

Endemic Species

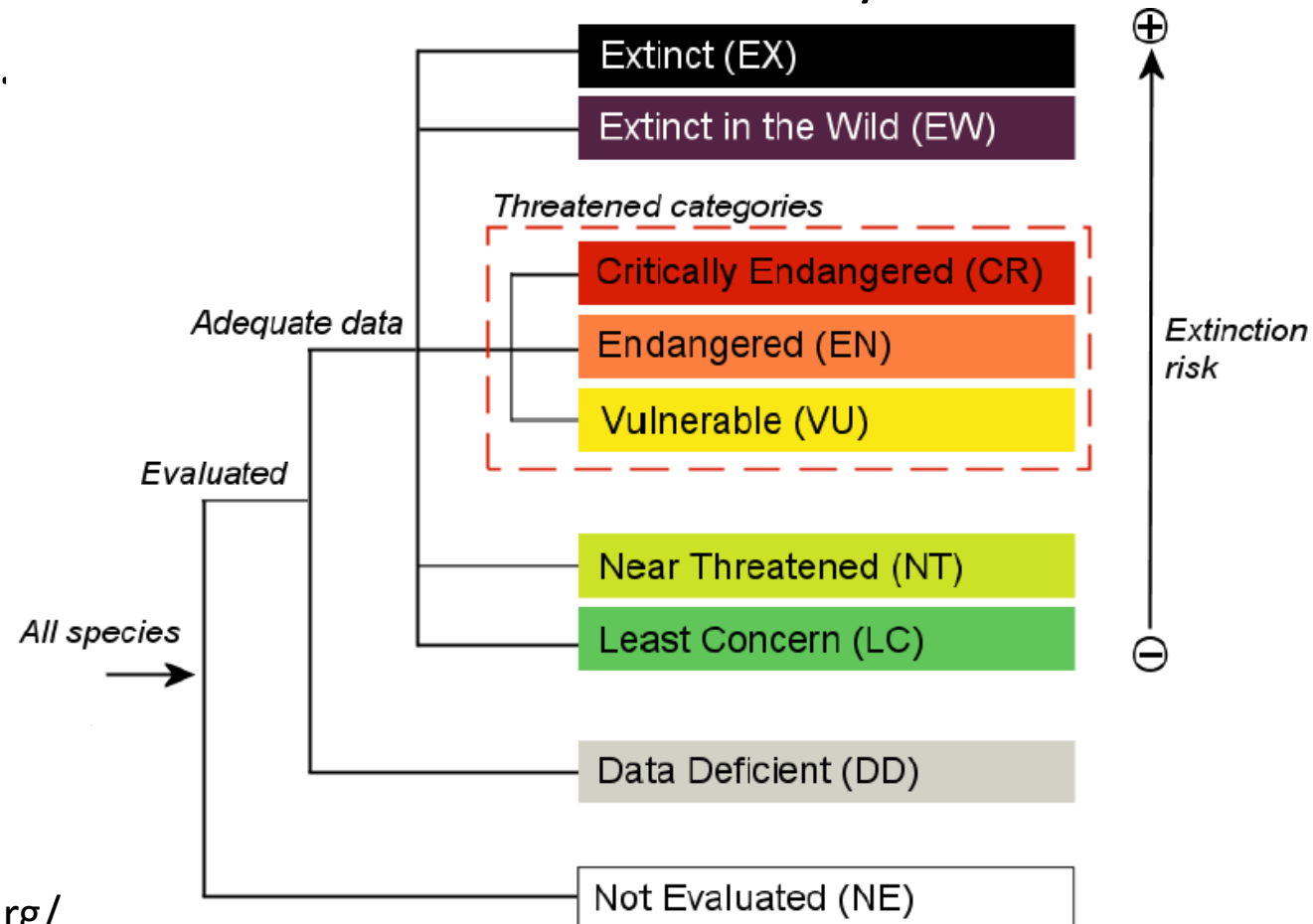


Michael P. Fogden/Bruce Coleman USA

This golden toad lived in Costa Rica's high-altitude Monteverde Cloud Forest Reserve. The species became extinct in 1989 apparently because its habitat dried up.

IUCN Red list

- IUCN (International Union for Conservation of Nature) evaluates the conservation status of plants and animal species.
- IUCN was set up in 1948 and the IUCN Red list system was first conceived in 1963.

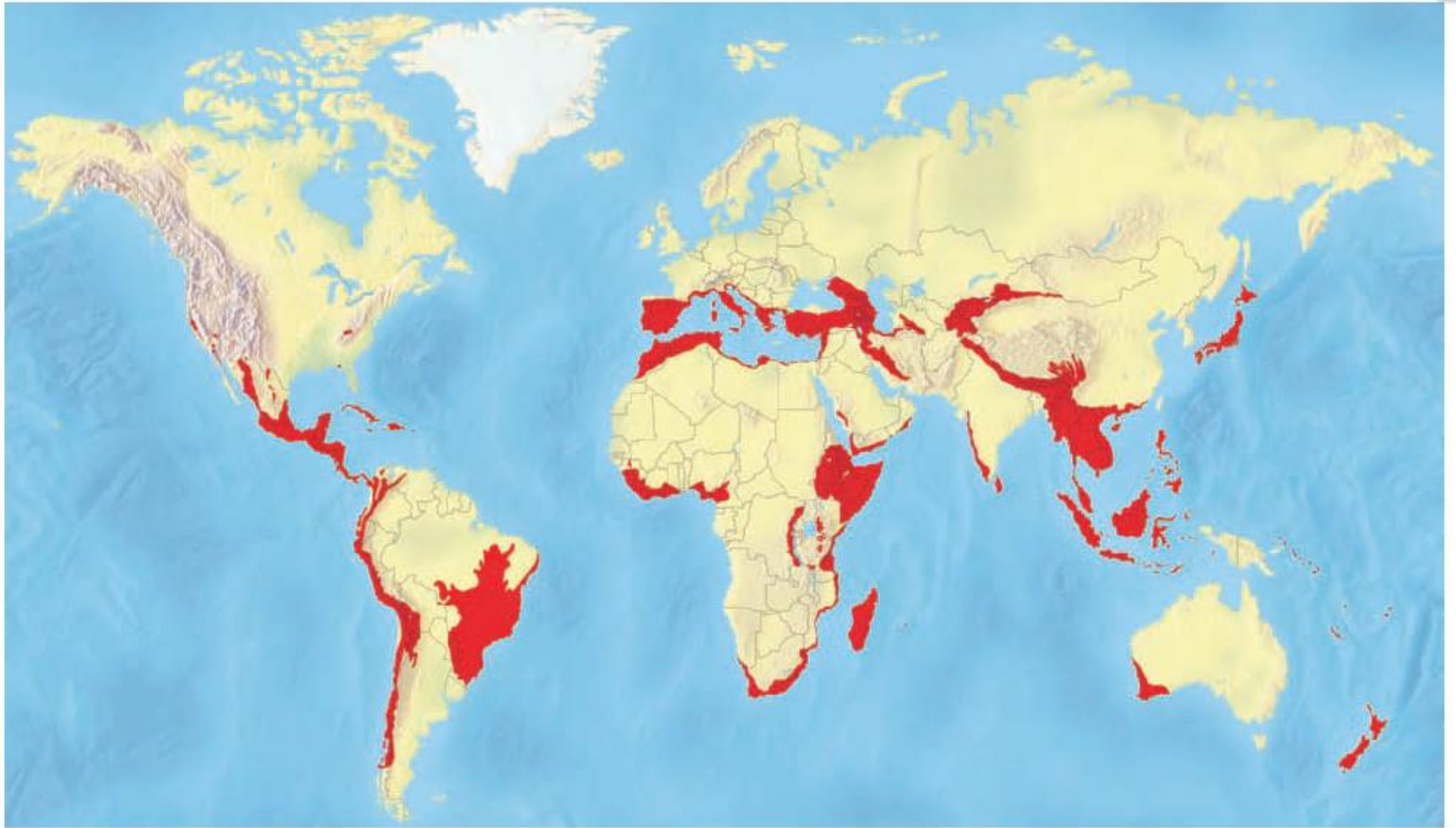


Hotspots

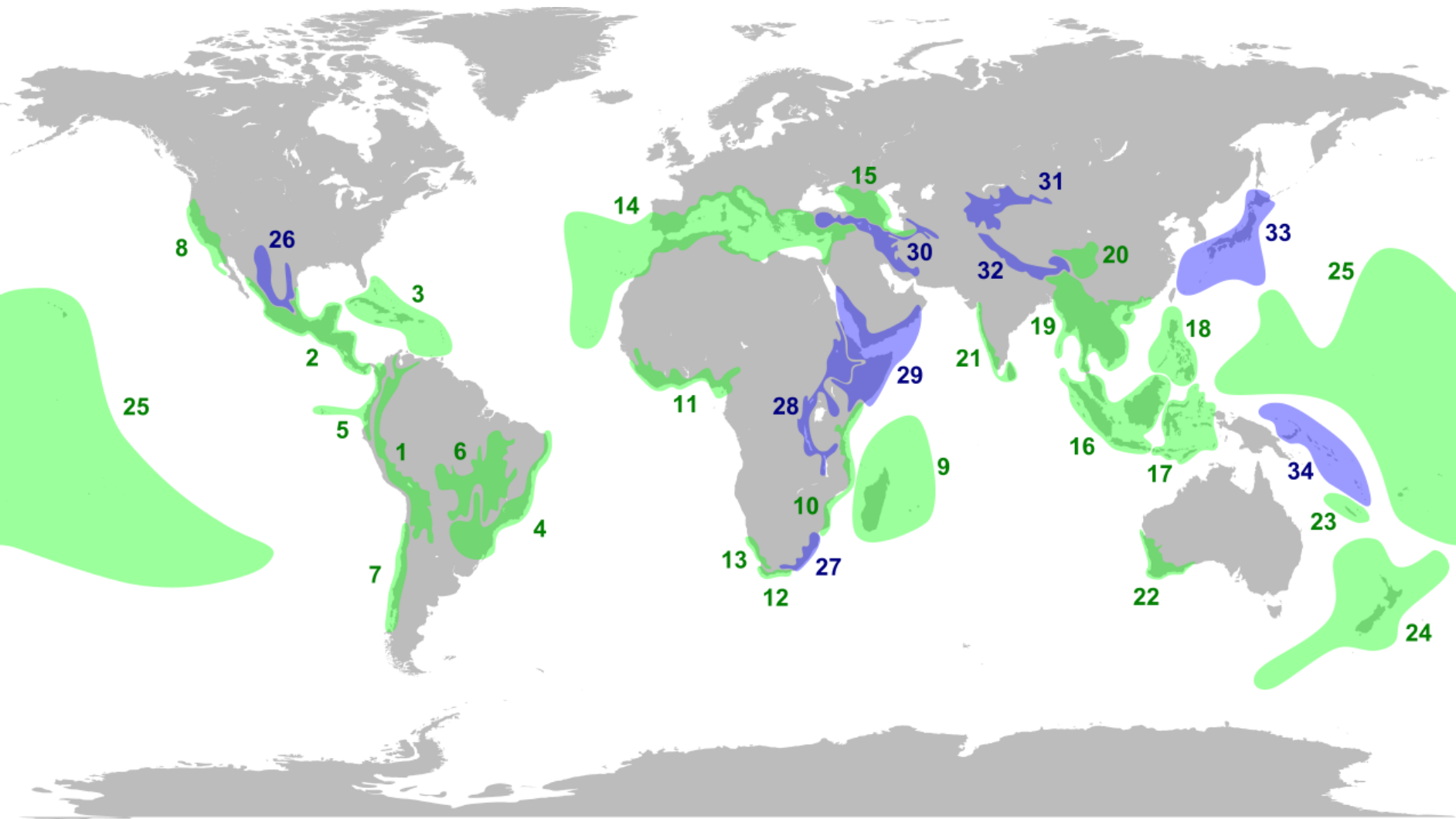
Biodiversity Hotspots

- Biogeographical region with significant levels of biodiversity that is ***threatened with destruction***.
- To qualify as a biodiversity hotspot, the region must
 - a) Contains at least **0.5% or 1500 species** of vascular plants as endemic, and
 - b) Has to have **lost** at least **70%** of its primary vegetation.
- Around the world, **36** areas considered as biodiversity hotspots.
- These 36 regions support nearly **60%** of the world's plant, bird, mammal, reptile and amphibian species.

Biodiversity Hotspots



36 Biodiversity Hotspots in World



*Original Proposal are in **green** and added regions are in **blue***

- **North and Central America – 5**

- California Floristic Province (8)
- Madrean pine-oak woodlands (26)
- Mesoamerica (2)
- Caribbean islands (3)
- North american coastal plane (36)

- **South America - 5**

- Atlantic forest (4)
- Cerrado (6)
- Chilean winter rainfall- valdivian forest (7)
- Tumbes-choco-magdalena (5)
- Tropical andes (1)

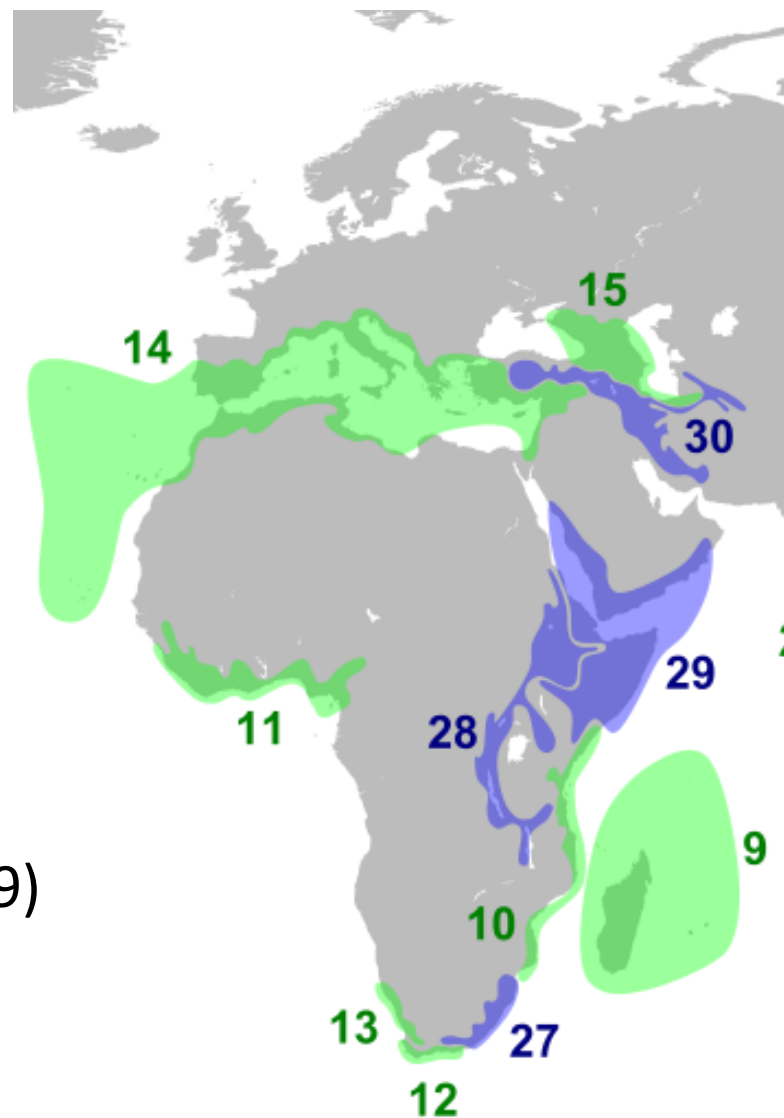


- **Europe - 1**

- Mediterranean basin (14)

- **Africa – 8**

- Cape floristic region (12)
- Coastal forests of eastern africa (10)
- Eastern afroalpine (28)
- Guinean forest of west africa (11)
- Horn of africa (29)
- Madagascar and Indian ocean island (9)
- Maputaland-pondoland-albany (27)
- Succulent Karoo (13)



- **Central Asia – 1**

- Mountains of central asia (31)

- **South Asia – 3**

- Eastern Himalaya (32)
- Indo-Burma, Myanmar (19)
- Western Ghats (21)
- Sri Lanka (21)

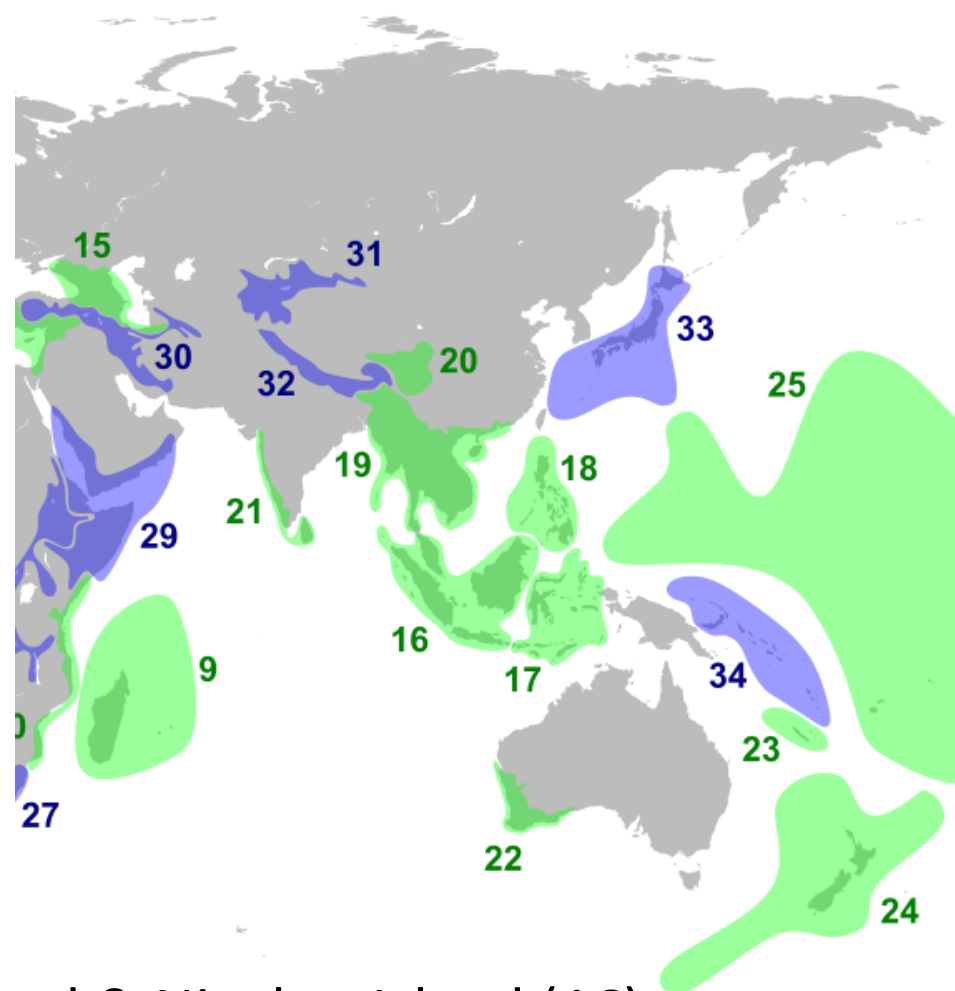
- **South East Asia, Asia Pacific –9**

- East Melanesian Island (34)
- New Caledonia (23)
- New Zealand (24)
- Sundaland & Nicobar Island (16)
- Philippines (18)
- Wallacea (17)

- Polynesia-Micronesia (25)
- Southwest Australia (22)
- Eastern Australian temperate forest

- **East Asia – 2** (Japan, Mt. Southwest china)

- **West Asia - 2** (Caucasus, Irano-Anatolian)

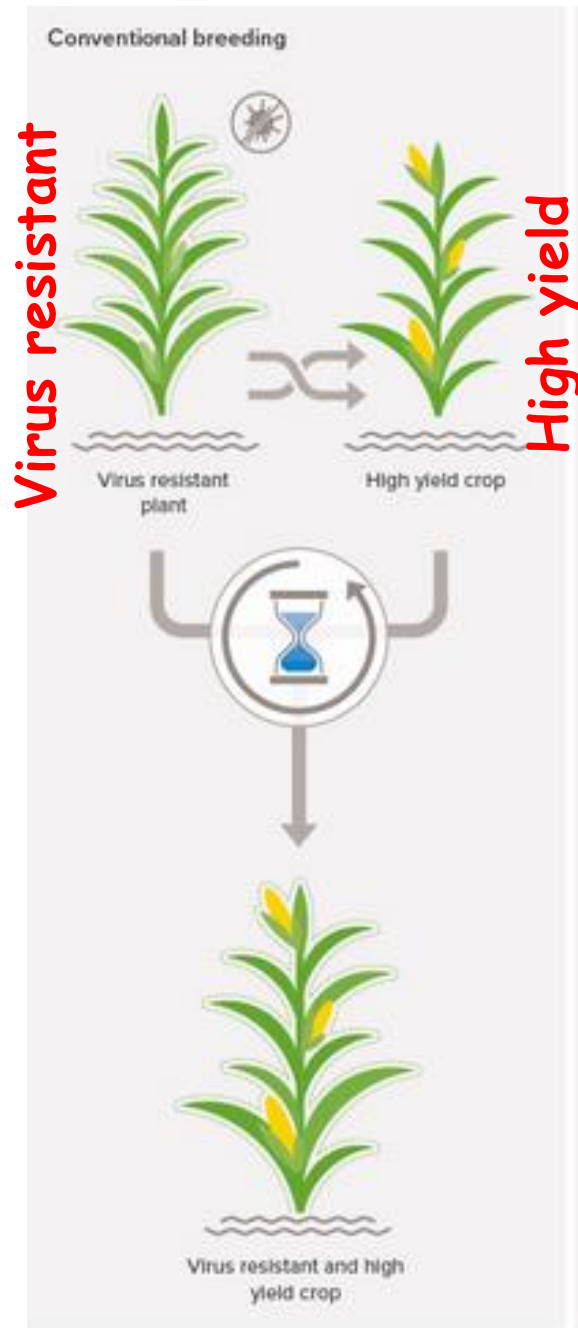


Biodiversity

- Importance
- types
- mega-biodiversity
- Species interaction
- Extinct, endangered and rare species;

Gene Modified Crops

Gene Modified Organisms



For centuries, farmers and scientists have used **crossbreeding** through artificial selection to develop genetically improved varieties of crops and livestock animals.

Limitation

- Traditional crossbreeding is a slow process, typically **taking 15 years** or more to produce a commercially valuable new crop variety,
- it can combine traits only from species that are genetically similar.

Gene Modified Organisms

scientists are creating a second *gene revolution* by using *genetic engineering* to develop *genetically improved* strains of crops and livestock animals

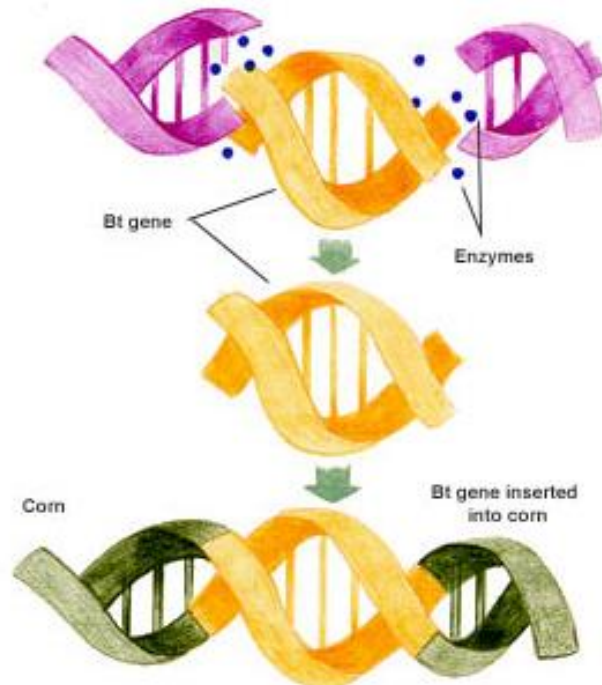
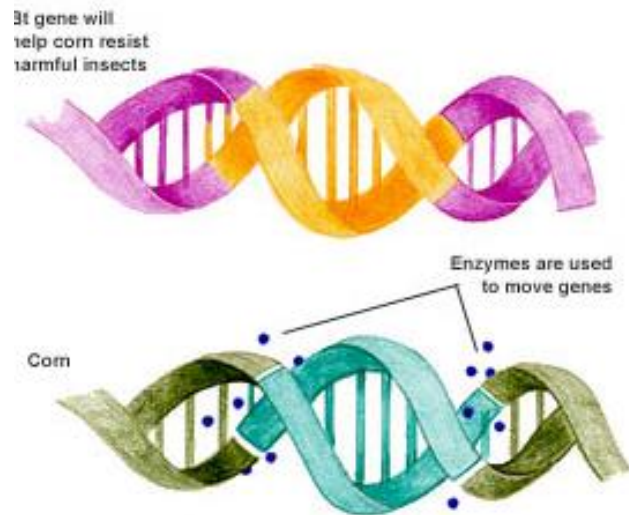
It involves altering an organism's genetic material through *adding, deleting, or changing segments* of its to produce desirable traits or to eliminate undesirable ones—a process called *gene splicing*.



Gene Modified Organisms

It involves altering an organism's genetic material through **adding, deleting, or changing segments** of its to produce desirable traits or to eliminate undesirable ones—a process called **gene splicing**.

It enables scientists to transfer genes between different species that would not normally interbreed in nature. The resulting organisms are called **genetically modified organisms (GMOs)**.



Gene Modified Organisms

Comparison with crossbreeding

- Compared to traditional crossbreeding, developing a new crop variety through gene splicing takes less time
- usually costs less
- Allows for the insertion of genes from almost any other organism into crop cells.

Gene Modified Organisms

Plans to develop

- Bioengineers plan to develop new GM varieties of crops that are resistant to heat, cold, herbicides, insect pests, parasites, viral diseases, drought, and salty or acidic soil.
- Several seed companies are in the final stages of offering a drought-tolerant GM version of corn.
- They also hope to develop crop plants that can grow faster and survive with little or no irrigation and with less fertilizer and pesticides.

GMO Foods

Tomato



Tomatoes have been genetically modified, but they are not being grown commercially at this time

Alfalfa



GMO alfalfa is contaminating non-GMO alfalfa crops at a rapid rate

Cotton



At least half of cotton grown in the world is GMO

Rice



GMO rice has been approved but is not yet being used commercially

Wheat



Unapproved GMO has contaminated wheat fields, and we don't yet know the extent of it

Sweet Corn



More than 70 percent of corn grown in the United States has been genetically engineered

Sugar Beets



90% of Sugar Beets (used to make 50% of our sugar) are GMO

Summer Squash



Farmers don't like GMO squash but some experts say GM squash have blended with wild squash

Salmon



GMO salmon has not been approved by the FDA, but it will be very soon

Soy



More than 93% of soybeans the United States produces are genetically modified

Canola Oil



87% of canola grown commercially, and 80% of wild canola is GMO

Peas



Peas have been genetically modified but are not approved or available

Yeast



GMO yeast for wine has been approved

Hawaiian Papaya



Most Hawaiian papaya is GMO, even many organic crops are contaminated

organic lifestyle
MAGAZINE

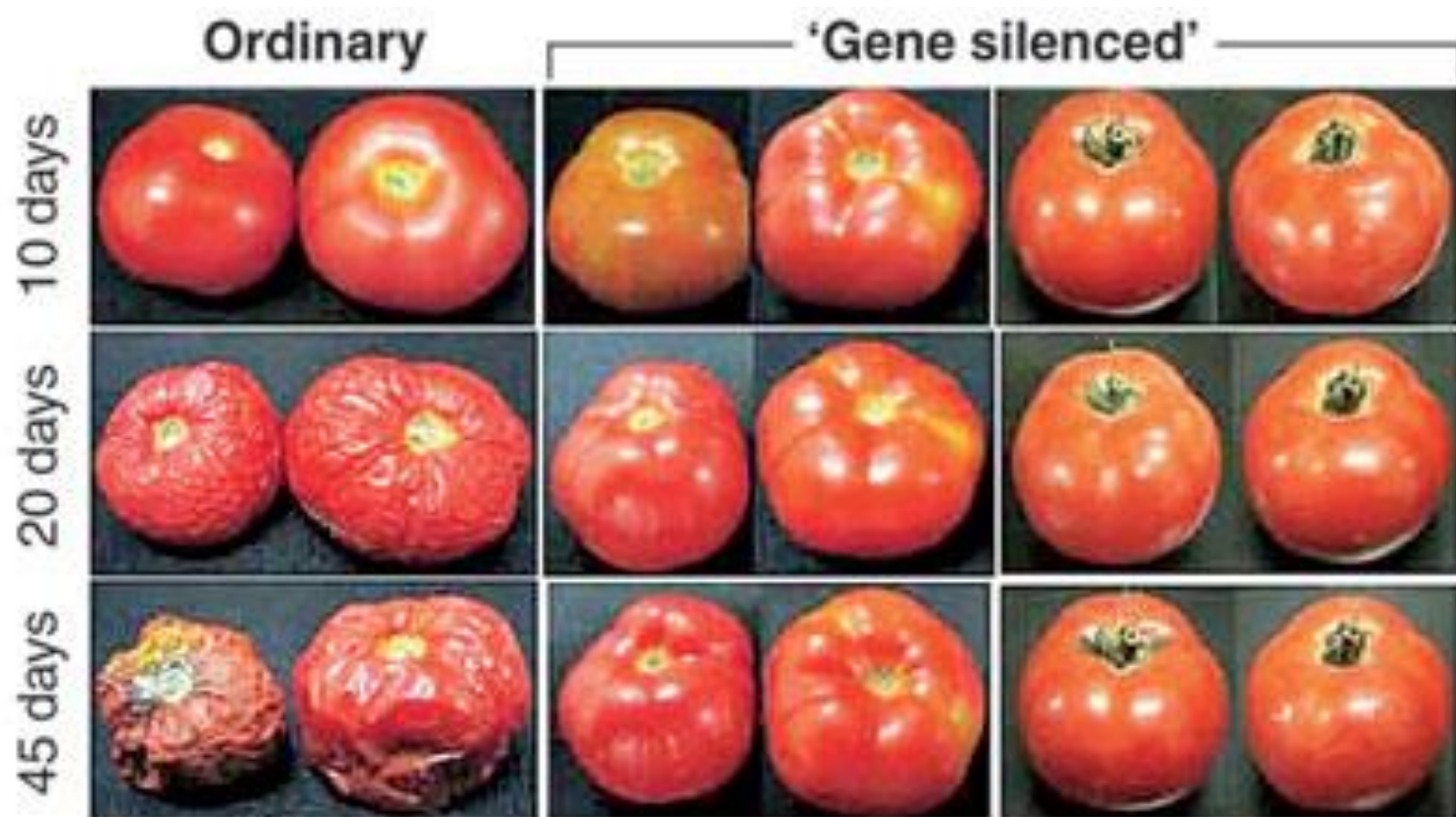


Image shows three sets of tomatoes. The ordinary control tomatoes (extreme left) soften and shrivel up, while texture of gene-silenced tomatoes remains intact for up to 45 days.

BT - Cotton

Bt Cotton: Produces Insecticide to cotton bollworm

Bt – *Bacillus thuringensis* – produces more than 200 different toxins, each harmful to different insects.

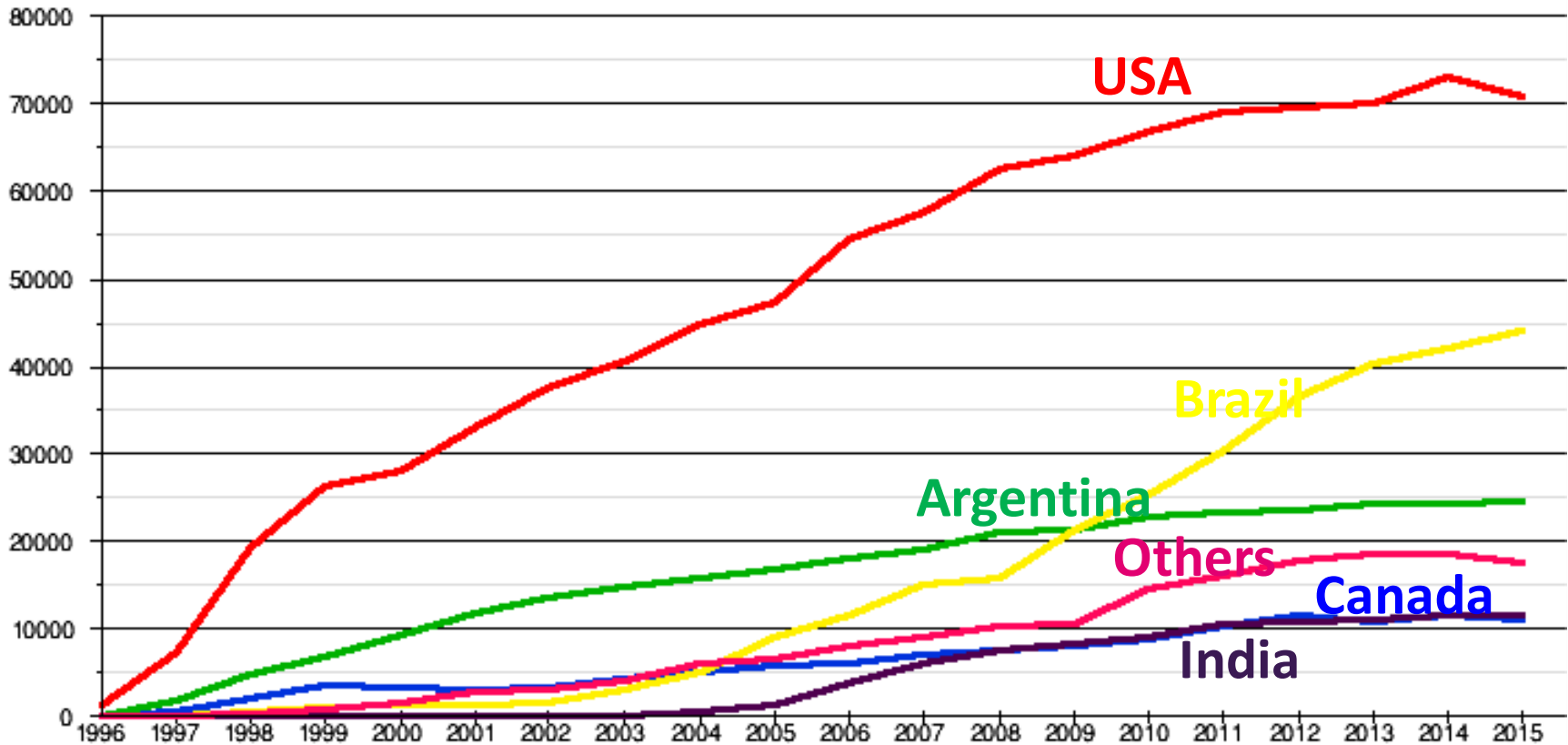
The gene coding for Bt toxin is inserted into cotton as a transgene, causing it to ***produce this natural insecticide*** in its tissues.

Introduced to India by the joint venture of Monsanto and Mahyco in 2002.

In 2011, India grew the largest GM cotton crop at 10.6 million hectares (US cotton crop was 4.0 million hectare)



Area planted in GM crops in 5 largest GM crop producing countries



Y-axis: ×1000 hectares

X-axis: Year

Advantages of GM Crops

- *Enhanced* desired traits
- Pest *resistance* and herbicide *tolerance*
- *Lesser time* for production than normal crops
- *Higher yield*
- *Reduced production cost*, required for chemical and mechanical needs in planting, maintenance
- *Tolerant* to environmental conditions

Disadvantages of GM Crops

- In 2009, biologist and genetic engineering expert Doug Gurian-Sherman pointed out that despite 20 years of research and 13 years of commercial sales, genetically engineered crops **have failed to significantly increase U.S. crop yields.**
- Critics also warn that we know too little about the **long-term potential harm** to human health and ecosystems from the widespread use of such crops.
- Critics point out that if GM organisms that are released into the environment cause some **unintended harmful genetic and ecological effects**, as some scientists expect, those organisms cannot be recalled.

Disadvantages of GM Crops

- *Harmful* to organisms and may cause health problems
- Does *not* taste “natural”
- Spread of *superweeds and superbugs*
- Larger companies have more power and possible greed to GMO manufactures (GM crops are a direct extension of chemical agriculture and are developed and sold by the world’s biggest chemical companies).
- Unforeseen *allergen risks*
- Long-term impact of GM crops are unknown.

Trade-Offs

Genetically Modified Crops and Foods

Advantages

Need less fertilizer

Need less water

More resistant to insects, disease, frost, and drought

Grow faster

May need less pesticides or tolerate higher levels of herbicides

May reduce energy needs



Disadvantages

Unpredictable genetic and ecological effects

Harmful toxins and new allergens in food

No increase in yields

More pesticide-resistant insects and herbicide-resistant weeds

Could disrupt seed market

Lower genetic diversity

Safety Issue of GM Crops

- In U.S., the government has approved GM crops based on their experimental studies.
- Most of the developed nations ***do not consider GM Crops to be safe***. In more than 60 countries around the world, including Australia, Japan and countries in European union, there are significant restrictions and outright bans on the production and sale of GM crops.

Ecological Succession

Ecological Succession

The types and numbers of species in biological communities and ecosystems change in response to wildfires,
volcanic eruptions,
climate change,
clearing of forests to plant crops.

The structure and species composition of communities and ecosystems change in response to changing environmental conditions through a process called **ecological succession**.

Ecological Succession

Ecologists recognize two main types of ecological succession, depending on the conditions present at the beginning of the Process

Primary ecological succession

secondary ecological succession

Ecological Succession

Primary ecological succession

Involves the gradual establishment of biotic communities in lifeless areas where there is no soil in a terrestrial ecosystem or no bottom sediment in an aquatic ecosystem.

Examples:

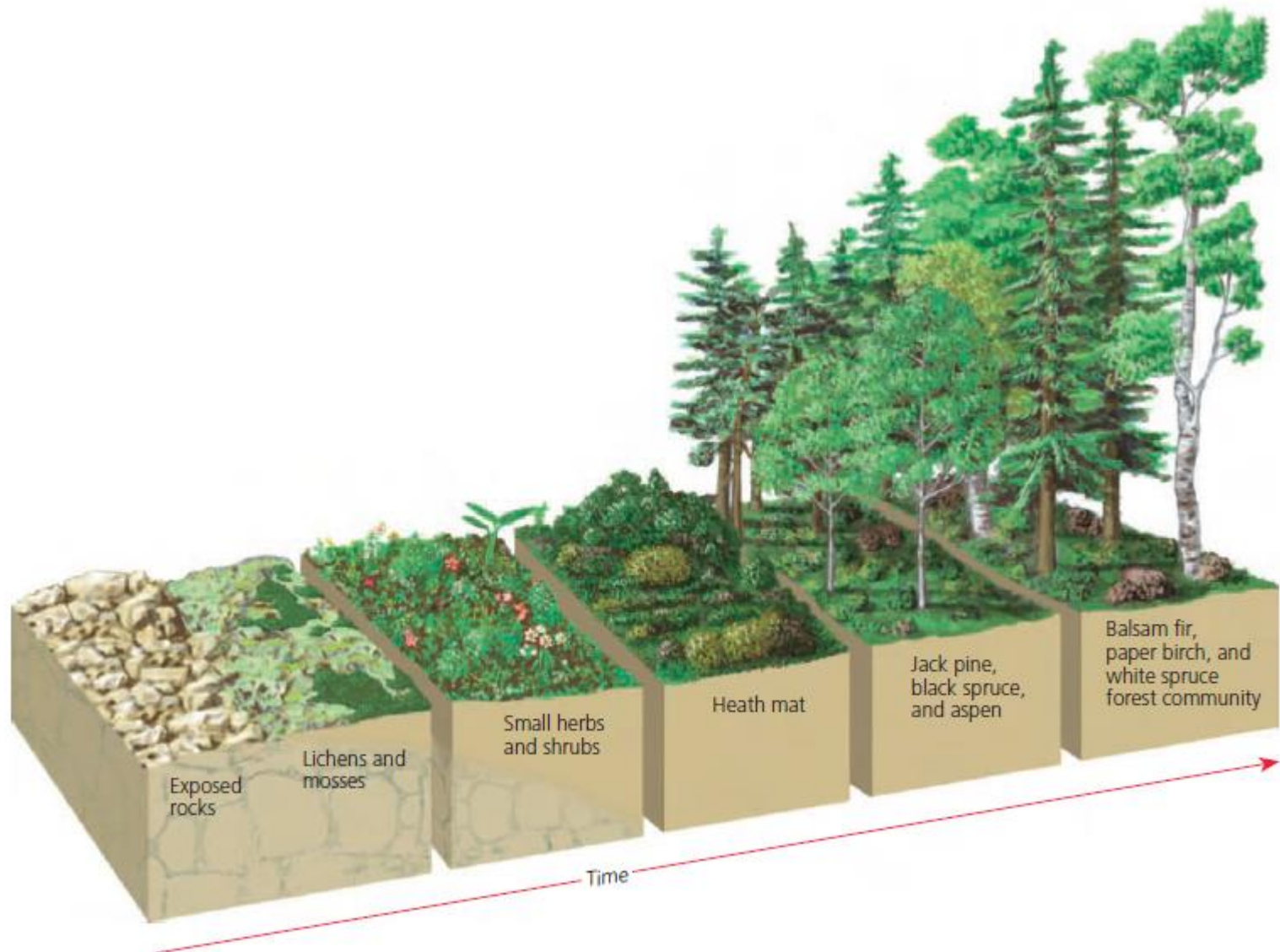
bare rock exposed by a retreating glacier,

newly cooled lava,

an abandoned highway or parking lot,

Ecological Succession

Primary ecological succession



Ecological Succession

Primary ecological succession

How long does it take for Primary ecological succession ??

Primary succession usually takes **hundreds to thousands of years** because of the need to build up fertile soil or aquatic sediments to provide the nutrients needed to establish a plant community.

Ecological Succession

Secondary ecological succession

- In which a series of communities or ecosystems with different species develop in places containing soil or bottom sediment.
- This type of succession begins in an area where an ecosystem has been disturbed, removed, or destroyed, but some soil or bottom sediment remains.

Examples:

Abandoned farmland,

burned or cut forests,

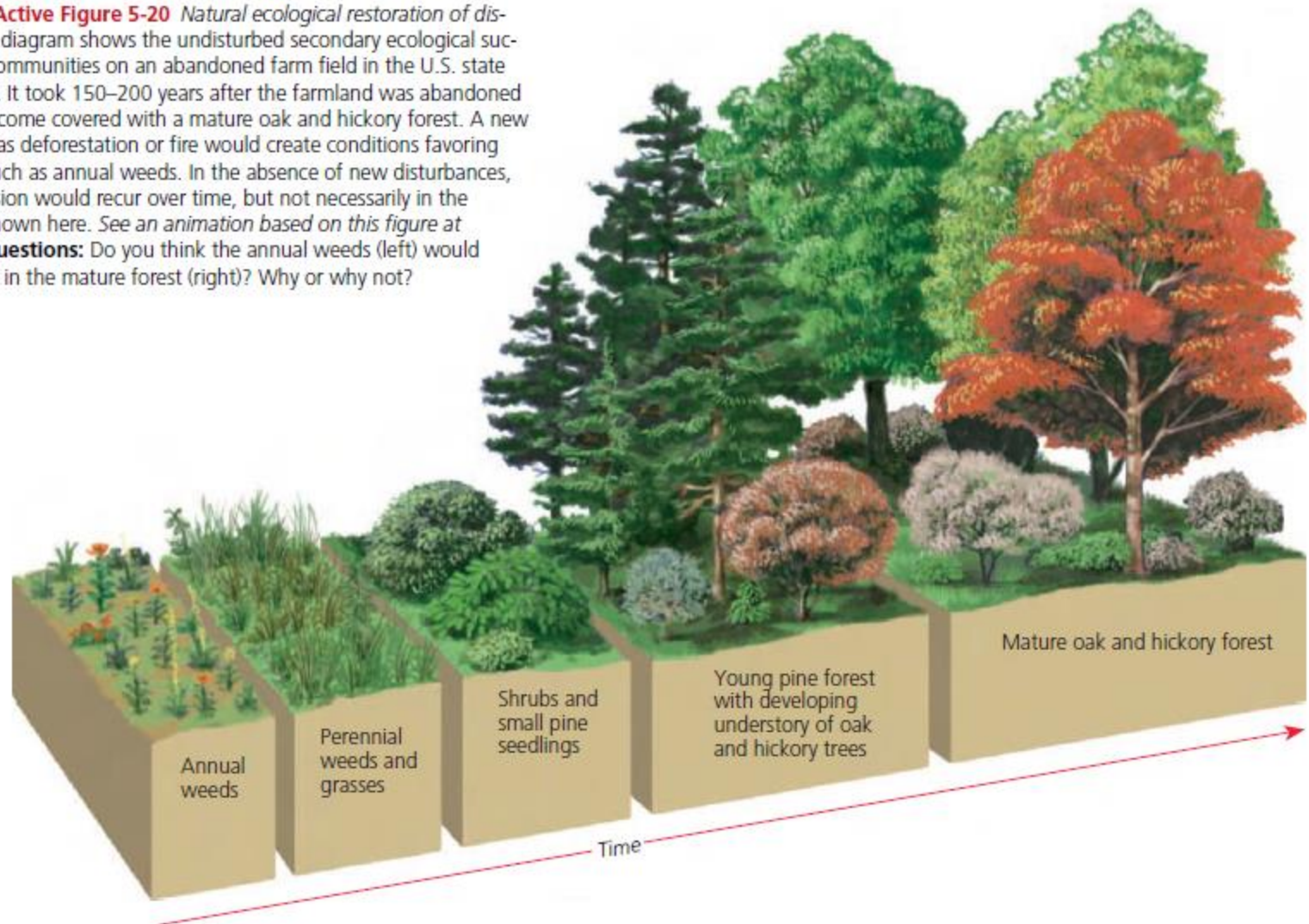
heavily polluted streams,

Land that has been flooded.

Ecological Succession

Secondary ecological succession

CENGAGENOW® **Active Figure 5-20** *Natural ecological restoration of disturbed land:* This diagram shows the undisturbed secondary succession of plant communities on an abandoned farm field in the U.S. state of North Carolina. It took 150–200 years after the farmland was abandoned for the area to become covered with a mature oak and hickory forest. A new disturbance such as deforestation or fire would create conditions favoring pioneer species such as annual weeds. In the absence of new disturbances, secondary succession would recur over time, but not necessarily in the same sequence shown here. See an animation based on this figure at CengageNOW. **Questions:** Do you think the annual weeds (left) would continue to thrive in the mature forest (right)? Why or why not?



Ecological Succession

Secondary ecological succession

How long does it take for Secondary ecological succession ??

Because some soil or sediment is present, new vegetation can begin to germinate, usually **within a few weeks**.

It begins with seeds already in the soil and seeds imported by wind or in the droppings of birds and other animals.

Ecological Succession

Importance

Primary and secondary ecological succession are important natural services that tend to **increase biodiversity**,

- species richness and interactions among species
- promoting population control
- enhances the energy flow and nutrient cycling
- sustainability of communities and ecosystems

As part of the earth's natural capital, both types of succession are examples of **natural ecological restoration**.

Threats to biodiversity

Threats to biodiversity

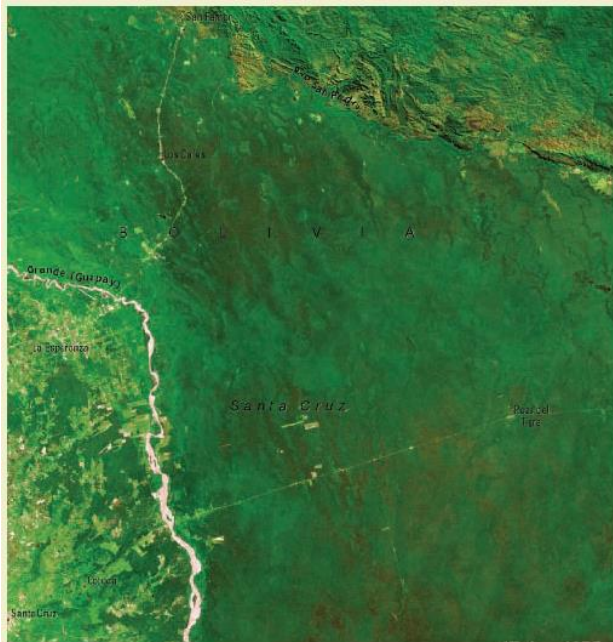
- Habitat loss
- Habitat degradation and fragmentation
- Introduction of non-native species
- Commercial hunting and poaching
- Sale of exotic pets and decorative plants
- Predator and pest control
- Pollution
- Climate change
- Overfishing

Threats to biodiversity

Loss of Habitat Is the Single Greatest Threat to Species:

According to biodiversity researchers, the greatest threat to wild species is habitat loss, degradation, and fragmentation.

Bolivian city of Santa Cruz



1975



2003

Threats to biodiversity

Habitat fragmentation occurs when a large, intact area of habitat such as a forest or natural grassland is divided, typically by roads, logging operations, crop fields, and urban development, into smaller, isolated patches or "habitat islands"

affect

It can divide populations of a species into smaller, increasingly isolated groups that are more vulnerable

Threats to biodiversity

Loss of Habitat Is the Single Greatest Threat to Species:

- Deforestation in tropical areas
- destruction and degradation of coral reefs
- Degradation of coastal wetlands,
- the plowing of grasslands
- the pollution of streams, lakes, and oceans.



Threats to biodiversity

Some Deliberately Introduced Species Can Disrupt Ecosystems

introduced species such as corn, wheat, rice, and other food crops, as well as species of cattle, poultry, and other livestock

affect

In their new habitats, some introduced species face no natural predators, competitors, parasites, or pathogens that would help to control their numbers in their original habitats.

Such non-native species can thus crowd out populations of many native species, trigger ecological disruptions, cause human health problems

Threats to biodiversity

Some Accidentally Introduced Species Can Disrupt Ecosystems

- Many unwanted non-native invaders arrive from other continents as stowaways on aircraft, in the ballast water of tankers and cargo ships

Affects

- Many tourists return home with living plants that can multiply **and become invasive**.
- Some of these plants might also contain insects that can escape, **multiply rapidly, and threaten crops**.

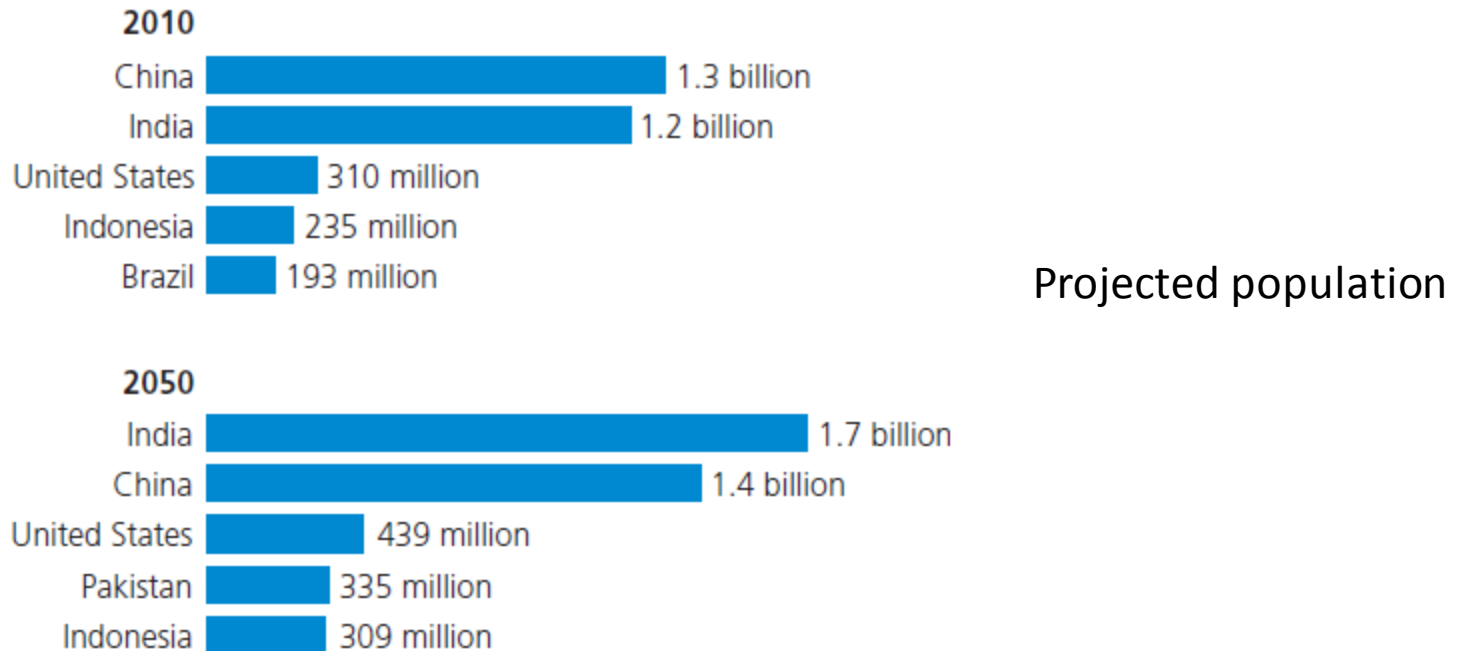
What Can You Do?

Controlling Invasive Species

- Do not capture or buy wild plants and animals.
- Do not remove wild plants from their natural areas.
- Do not release wild pets back into nature.
- Do not dump the contents of an aquarium into waterways, wetlands, or storm drains.
- When camping, use wood found near your campsite instead of bringing firewood from somewhere else.
- Do not dump unused bait into waterways.
- After dogs visit woods or the water, brush them before taking them home.
- After each use, clean your mountain bike, canoe, boat, motor, and trailer, all fishing tackle, hiking boots, and other gear before heading for home.

Threats to biodiversity

Population Growth and Overconsumption



human population growth and excessive and wasteful consumption of resources have greatly expanded the human ecological footprint,

ecological footprint has eliminated, degraded, and fragmented vast areas of wildlife habitat

Threats to biodiversity

- Pollution

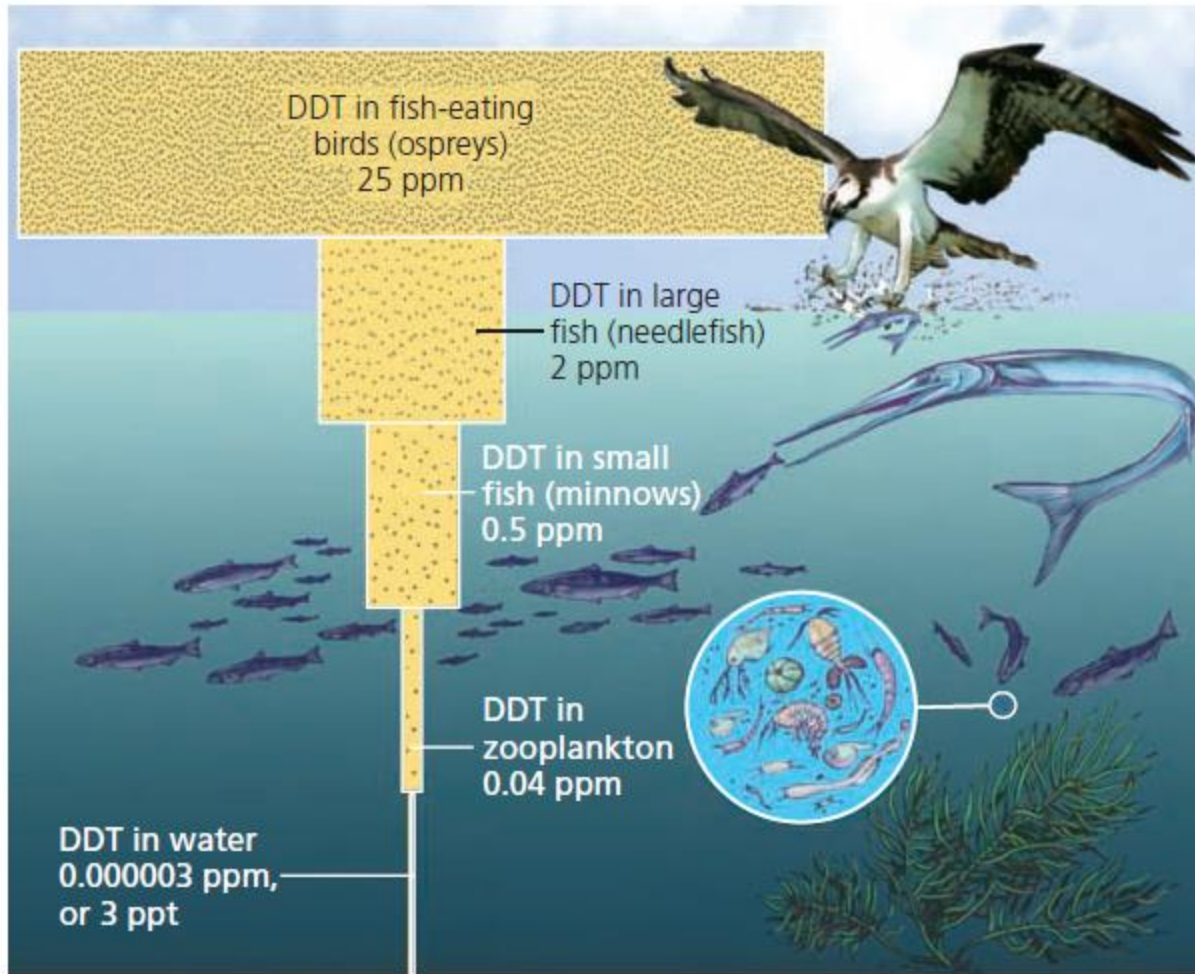
- Pollution also threatens some species with extinction as has been shown by the unintended effects of certain pesticides.

Examples:

- pesticides kill about one fifth of the honeybee colonies that pollinate almost a third of U.S. food crops
- They also kill more than 67 million birds and 6-14 million fish each year, and they threaten about one-fifth of the country's endangered and threatened species.
- During the 1950s and 1960s, populations of fish eating birds such as ospreys, brown pelicans, and bald eagles plummeted. A chemical derived from the pesticide DDT, entered surface waters

Threats to biodiversity

- Pollution



Threats to biodiversity

- Climate change

projected *climate change* could help to drive a quarter to half of all land animals and plants to extinction by the end of this century.

Examples:

- Scientific studies indicate that **polar bears** and 10 of the world's 17 **penguin** species are already threatened because of higher temperatures and melting sea ice in their polar habitats.

Threats to biodiversity

- Climate change

projected *climate change* could help to drive a quarter to half of all land animals and plants to extinction by the end of this century.



Threats to biodiversity

- Illegally Killing, Capturing and Selling of Wild Species

Some protected species are illegally killed (poached) for their valuable parts or are sold live to collectors.

- the Indian or Bengal tiger is at risk because a coat made from its fur can sell for as much as \$100,000 in Tokyo.



Threats to biodiversity

- Illegally Killing, Capturing and Selling of Wild Species

Some exotic plants, especially orchids and cacti are endangered because they are gathered (often illegally) and sold to collectors to decorate houses, offices, and landscapes.



Threats to biodiversity

- Illegally Killing, Capturing and Selling of Wild Species

Overexploitation of birds and other species is a major threat to bird populations

- Fifty-two of the world's 388 parrot species are threatened partly because so many parrots are captured (often illegally) for the pet trade.



Threats to biodiversity

- Overfishing

- Modern industrial fishing has caused as much as 80% depletion of some wild fish species in only 10-15 years.
- Overfishing usually results in only a temporary depletion of fish stocks, as long as depleted areas and fisheries are allowed to recover.



Conservation of Biodiversity

Conservation of Biodiversity

Some of the remedial measures mentioned below will be useful for conservation and restoration of the biotic resources

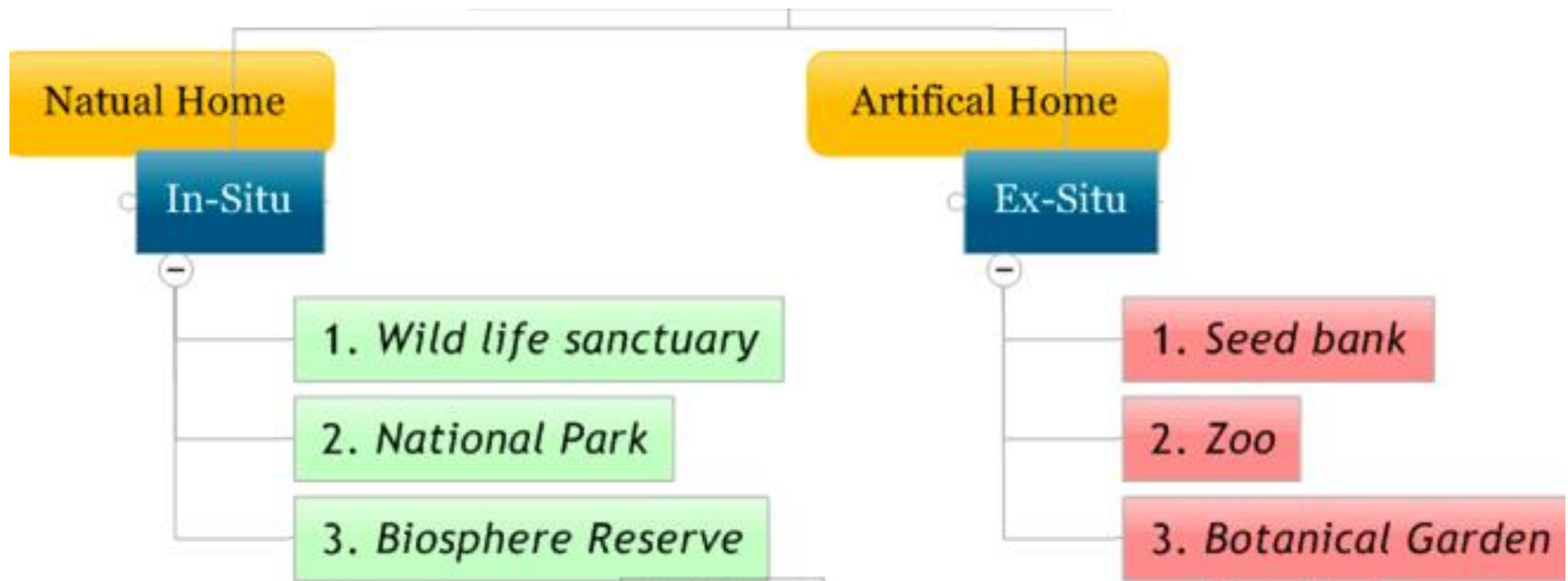
- **Assessment** of the biodiversity inventory
- Wildlife conservation measures focussing on **protecting** animal and plant life in zoos, sanctuaries, national parks and gardens.
- Setting up of **seed bank and gene bank**
- Modification of agricultural practices encouraging mixed cropping and poly culture

- ***Restoration*** of habitats and eco-systems which are important for biological communities (Reforestation, prevention of soil erosion, fencing, fertilisation).
- Imparting environmental ***education and awareness*** and motivate communities to conserve resources
- Introduction and implementation of stringent legislations
- ***Population control*** and check on indiscriminate urbanization.

Conservation Approach

Two basic approaches of conservation

- **In situ conservation:** Conserving animals and plants in their natural habitats. Large areas are kept reserve for wild life
- **Ex situ conservation:** Conservation is done under human care. Endangered plants and animals are collected and are cared for and reproduced in controlled conditions.



In Situ Conservation

Wildlife Sanctuary	Biosphere Reserves	National Parks
It is a natural habitat, owned by the government or private agency, that safeguards a particular species of birds and animals	A reserved area of land established by the government to protect the environment as a whole	A national park is a reserved area of land owned by the government which is protected from industrialization, human exploitation, and pollution. National Parks may cover Sanctuaries, and reserves as well.
Animals, birds, insects, reptiles, etc	Meant to preserve the biodiversity of a specified area	Flora, fauna, landscape, historical objects, etc

In Situ Conservation in India

- There are **18** biosphere reserves, **103** national parks and **543** wildlife sanctuaries in India.
- **Biosphere Reserves**: Protects larger areas of natural habitat, than a national park or animal sanctuary and often include one or more national parks
- **National Park** is an area dedicated for the conservation of wildlife along with its environment. Each National Park usually aims at conservation specifically of some particular species of wildlife along with others.
- **Wildlife Sanctuaries**: also protects the wildlife from killing, hunting, shooting etc., by humans.

Ex Situ Conservation

- This type is mainly for conservation of crop varieties and all the local varieties with main objective of conserving the total genetic variability of the crop species for ***future crop improvement***.

Important seed/gene bank in india are

- National Bureau of Plant Genetic Resources (**NBPGR**) in New Delhi
- National Bureau of Animal Genetic Resources (**NBAGR**) in Karnal, Haryana
- National Facility for Plant Tissue Culture Repository (**NFPTCR**) created within NBPGR.
- There are about 40 Botanical gardens available in India

Ex Situ Conservation

Zoological parks	Botanical gardens
It is a place where wild animals are kept under human observation in the natural environment.	A botanical garden is defined as that place, where different varieties of plants are cultivated.
The main purpose of setting zoological parks are to maintain ecological balance in the environment and conserve wildlife.	Botanical gardens are established for conservation of flora and scientific study.

How can we protect wild species from extinction?

What Can You Do?

Protecting Species

- Do not buy furs, ivory products, or other items made from endangered or threatened animal species.
- Do not buy wood or paper products produced by cutting old-growth forests in the tropics.
- Do not buy birds, snakes, turtles, tropical fish, and other animals that are taken from the wild.
- Do not buy orchids, cacti, or other plants that are taken from the wild.
- Spread the word. Talk to your friends and relatives about this problem and what they can do about it.

Conservation: Terrestrial
and aquatic biodiversity

Conservation of Terrestrial biodiversity

- sustain forests
- sustain grasslands
- sustain parks and nature reserves

sustain forests

Manage and Sustain Forests

Solutions

More Sustainable Forestry

- Identify and protect forest areas high in biodiversity
- Rely more on selective cutting and strip cutting
- Stop clear-cutting on steep slopes
- Stop logging in old-growth forests
- Sharply reduce road building in uncut forest areas
- Leave most standing dead trees and fallen timber for wildlife habitat and nutrient cycling
- Put tree plantations only on deforested and degraded land
- Certify timber grown by sustainable methods
- Include ecological services of forests in estimates of their economic value

Manage and Sustain Forests

we can improve the management of forest fires

severe fires could threaten forest lands

Increased logging left behind highly flammable logging debris (called *slash*), and greater public use of federal forest lands also contribute to this fire danger.

Manage and Sustain Forests

We can reduce the demand for harvested trees

- One reason for cutting trees is to provide pulp for **making paper**, but paper can be made from fiber that does not come from trees.
- A 1987 study by a Canadian paper mill found that newspaper pages made from **kenaf fiber** were brighter and stronger, and had less environmental impact than newspaper made from pine fibers.

Manage and Sustain Forests

Governments and individuals can act to reduce tropical deforestation

Individual role

- Individuals can plant trees
- we can all reduce the demand for products that fuels illegal and unsustainable logging in tropical forests.
- We can also use substitutes for wood such as recycled plastic building materials and bamboo
- we can sharply reduce the use of throwaway paper products and replace them with reusable plates, cups, and cloth napkins and handkerchiefs

Solutions

Sustaining Tropical Forests

Prevention

Protect the most diverse and endangered areas

Educate settlers about sustainable agriculture and forestry

Subsidize only sustainable forest use

Protect forests through debt-for-nature swaps and conservation concessions

Certify sustainably grown timber

Reduce poverty

Slow population growth



Restoration

Encourage regrowth through secondary succession

Rehabilitate degraded areas

Concentrate farming and ranching in already-cleared areas

sustain grasslands



Manage and Sustain Grassland

After forests, grasslands are the ecosystems most widely used and altered by human activities.

Rangelands are unfenced grasslands in temperate and tropical climates that supply forage, or vegetation for **grazing** (grass-eating) and **browsing** (shrub-eating) animals.



shrub

Manage and Sustain Grassland

Some rangelands are overgrazed

- Overgrazing occurs when too many animals graze for too long, damaging the grasses and their roots, and exceeding the carrying capacity of a rangeland area



Manage and Sustain Grassland

We can manage rangelands more sustainably

- The most widely used method for more sustainable management of rangelands is to control the number of grazing animals and the duration of their grazing in a given area so the carrying capacity of the area is not exceeded
- Protecting overgrazed land from further grazing by moving livestock around and by fencing off these damaged areas eventually leads to their natural ecological restoration by ecological succession.
- Replanting severely degraded areas with native grass seeds and applying fertilizer can increase the growth of desirable vegetation and reduce soil erosion

sustain parks and nature reserves

sustain parks and nature reserves

National Parks Face Many Environmental Threats

- Today, more than 1,100 major national parks are located in more than 120 countries
- Parks in less-developed countries have the greatest biodiversity of all parks globally, but only about 1% of these parklands are protected.
- Not all park visitors understand the rules that protect species, and rangers have to spend an increasing amount of their time on law enforcement instead of on conservation management and education.

sustain parks and nature reserves

National Parks Face Many Environmental Threats

Solutions

National Parks

- Integrate plans for managing parks and nearby federal lands
- Add new parkland near threatened parks
- Buy private land inside parks
- Locate visitor parking outside parks and provide shuttle buses for people touring heavily used parks
- Increase federal funds for park maintenance and repairs
- Raise entry fees for visitors and use resulting funds for park management and maintenance
- Seek private donations for park maintenance and repairs
- Limit the number of visitors in crowded park areas
- Increase the number of park rangers and their pay
- Encourage volunteers to give visitor lectures and tours

sustain parks and nature reserves

Nature Reserves Occupy Only a Small Part of the Earth's Land

- Most ecologists and conservation biologists believe the best way to preserve biodiversity is to create a worldwide network of **large protected areas**.
- Most developers and resource extractors oppose protecting the earth's remaining undisturbed ecosystems. They contend that these areas might contain valuable resources that would add to current economic growth.

sustain parks and nature reserves

Nature Reserves Occupy Only a Small Part of the Earth's Land



The Silver Creek Nature Conservancy Preserve, a high-desert ecosystem near Sun Valley in the U.S. state of Idaho,

sustain parks and nature reserves

Designing and Connecting Nature Reserves

- When deciding on whether to recommend large- or medium-sized reserves, conservation biologists must carefully consider variations in the ecosystems of a specific area.
- Establishing protected habitat corridors between isolated reserves helps to support more species and allows migration by vertebrates that need large ranges.
- The creation of large reserves connected by corridors on an eco-regional scale is a grand goal of many conservation biologists.

Conservation of aquatic biodiversity

Conservation: aquatic biodiversity

What are the major threats to aquatic biodiversity?

- Human activities are destroying and degrading aquatic habitat
- Invasive species are degrading aquatic biodiversity
- Population growth and pollution can reduce aquatic biodiversity
- Climate change is a growing threat
- Overfishing

Conservation: aquatic biodiversity

Laws and treaties have protected some endangered and threatened marine species

- 1975 Convention on International Trade in Endangered Species (CITES),
- the 1979 Global Treaty on Migratory Species,
- the 1995 International Convention on Biological Diversity.

Conservation: aquatic biodiversity

Economic incentives can help sustain aquatic biodiversity

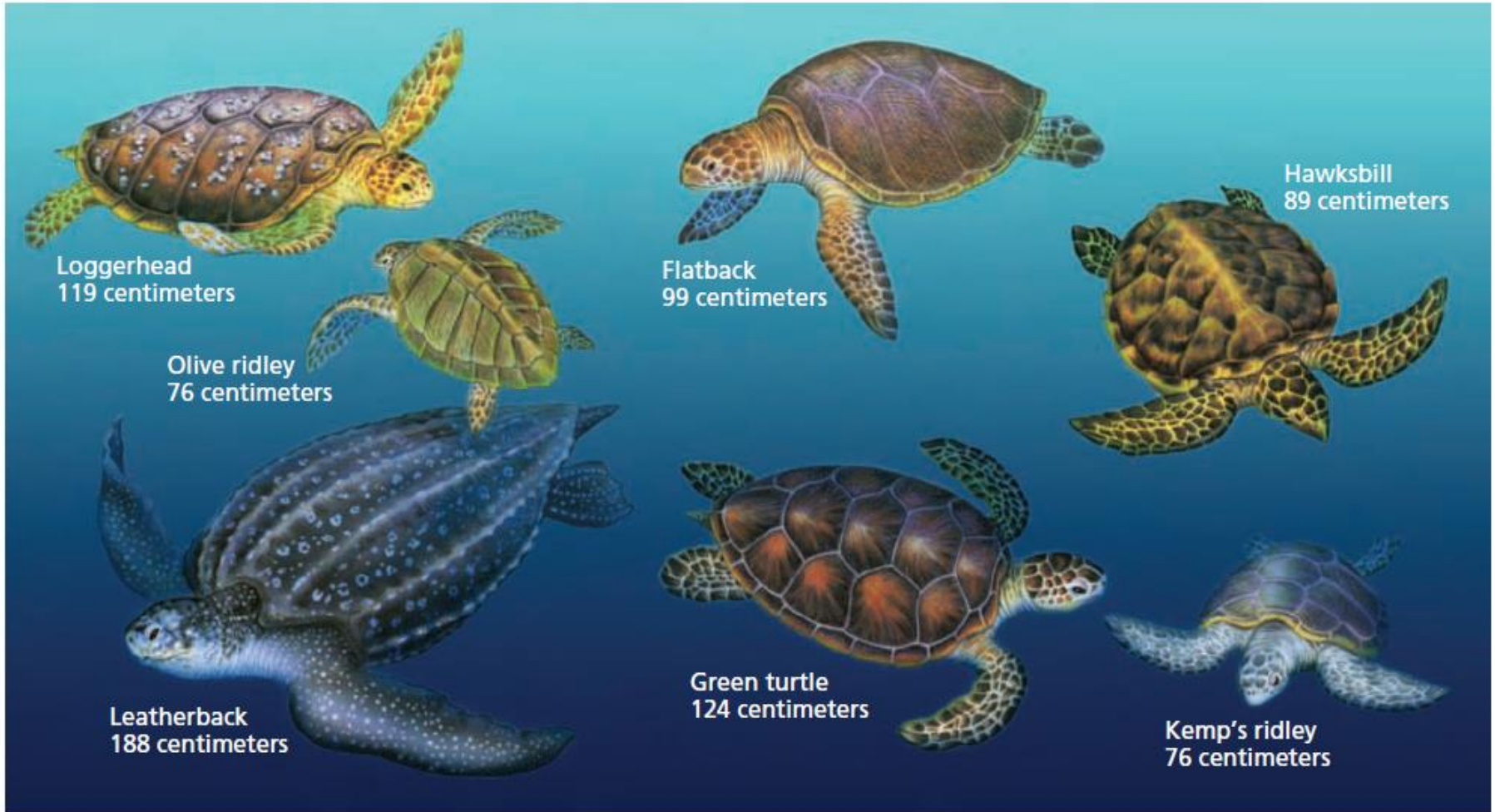
- Other ways to protect endangered and threatened aquatic species involve using economic incentives

Example:

- For example, according to a 2004 World Wildlife Fund study, sea turtles are worth more to local communities alive than dead.
- sea turtle tourism brings in almost three times more money than the sale of turtle products such as meat, leather, and eggs brings in

Conservation: aquatic biodiversity

Economic incentives can help sustain aquatic biodiversity



Conservation: aquatic biodiversity

Marine sanctuaries protect ecosystems and species

- Some countries are attempting to protect marine biodiversity and to sustain fisheries by establishing marine sanctuaries.
- Since 1986, the IUCN has helped to establish a global system of marine protected areas (MPAs)—areas of ocean partially protected from human activities.
- There are more than 4,000 MPAs worldwide.

Conservation: aquatic biodiversity

Establishing a global network of marine reserves: an ecosystem approach to marine sustainability

- The primary objective of this ecosystem approach is to **protect and sustain the whole marine ecosystem** for current and future generations instead of focusing primarily on protecting individual species.
- The foundation of this ecological approach is to establish a global network of fully **protected marine reserves**, some of which already exist.
- Such reserves would be closed to activities such as commercial fishing, dredging, and mining, as well as to waste disposal.

Conservation: aquatic biodiversity

Establishing a global network of marine reserves: an ecosystem approach to marine sustainability

- Example

In 2009, a group of western Pacific island nations, seeking to save the world's last great tuna stocks, put four areas of international waters off limits to fishing, creating the largest protected marine reserve in history.

Conservation: aquatic biodiversity

Protecting marine biodiversity requires commitments from individuals and communities

- Coastal residents must also think carefully about the chemicals they put on their lawns and the kinds of waste they generate.
- To deal with problems of pollution and overfishing, communities must closely monitor and regulate fishing and coastal land development, and greatly reduce pollution from land-based activities.

Conservation: aquatic biodiversity

Protecting marine biodiversity requires commitments from individuals and communities

- More important, each of us can make careful choices in purchasing only sustainably harvested and farmed seafood.
- Individuals can also reduce their carbon footprints in order to slow projected climate change and its numerous harmful effects on marine and other ecosystems,